Chromatic Admissible Contexts and Enforced Completeness of Nonadjacent Vertices

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In a d-uniform quantum hypergraph, every measurement context is rainbow in every strong d-coloring. We study *chromatic admissible contexts*: sets of d pairwise nonadjacent vertices that receive every color exactly once in all such colorings. Equivalently, their state sum is one on all extendable two-valued states. The stronger notion of a *two-valued state admissibility context* requires the same normalization on every two-valued state. We construct a faithful 53-ray, 34-context example in $\mathbb{R}^3$ by pasting two nonseparating Kochen–Specker gadgets at one ray. Its two independent admissible triples are normalized on all 224 two-valued states and are rainbow in all 768 labeled strong 3-colorings. A short implication proof establishes this universal normalization, whereas an explicit Born preparation gives total 8/9 on one triple. Admissibility is therefore distinct from affine and operator normalization. It also differs from Navara–Svozil pseudocontexts, whose balanced affine certificates equate two probability sums and imply color-multiset equivalence. Within the examined Yu–Oh restrictions, independent chromatic admissible triples first occur at $H_{20,12}$, with two triples; $H_{21,13}$ has 21, classified through its two color signatures. Structural color propagation proves universal completeness of its hub, core, and coplanar triples. Cascaded roof-twin gadgets $G_{41,25}$ and $G_{41,26}$ give further two-valued admissibility contexts, with a two-step binary proof and separately verified faithful coordinates. Explicit binary filters reduce the state spaces of $H_{20,12}$ and $H_{21,13}$ to their extendable states. Further real and complex examples exhibit universal rainbow subsets, chromatic twins, and color-multiset-equivalent pairs. A census of 19 faithfully represented configurations separates these assignment-level relations from affine conservation laws and identifies the state family on which admissibility holds.

I. INTRODUCTION

A complete rank-one projective measurement in $\mathbb{F}^d$ ($\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$) is represented by an orthonormal basis of rays $\{[q_v] : v \in e\}$ whose rank-one projectors $P_v = q_v q_v^\dagger / \|q_v\|^2$ resolve the identity,

$$\sum_{v \in e} P_v = I_d. \tag{1}$$

For any finite d-uniform hypergraph $H = (V, E)$ of intertwined projective measurements ($n = |E|$), every admissible probability assignment $p : V \to [0, 1]$—whether generated by the quantum Born rule $p(v) = \mathrm{Tr}(\rho P_v)$, by convex combinations of classical two-valued states $s : V \to \{0, 1\}$ in $\mathcal{S}(H)$, or by more general normalized weight assignments—satisfies the context normalization equations

$$\sum_{v \in e} p(v) = 1 \qquad (\forall\, e \in E). \tag{2}$$

We write $\mathcal{W}(H)$ for all normalized weights $p : V \to [0, 1]$ satisfying Eq. (2), and $\mathcal{S}(H) = \mathcal{W}(H) \cap \{0, 1\}^V$ for its two-valued states. The entire real affine solution space is

$$\mathcal{A}(H) = \Big\{x \in \mathbb{R}^V : \sum_{v \in e} x(v) = 1 \text{ for all } e \in E\Big\}. \tag{3}$$

These three domains must be distinguished: an identity on $\mathcal{S}(H)$ need not be an identity on $\mathcal{W}(H)$ or on $\mathcal{A}(H)$.

In Refs. [1, 2], Navara and Svozil introduced *pseudocontexts* (or *pseudoblocks*). Under the restrictive affine definition of the latter work, these are disjoint sets $A, B \subseteq V$ of equal size $k \geq 2$, each independent in the two-section graph $\Gamma(H)$, whose probability sums are forced to coincide by a balanced linear combination of context equations. In particular,

$$\sum_{a \in A} p(a) = \sum_{b \in B} p(b) \qquad (\forall\, p \in \mathcal{A}(H)). \tag{4}$$

The common sum need not equal one, and neither set is required to be complete or exclusive under a coloring. When H admits a faithful orthogonal representation (FOR) [3, 4] in $\mathbb{F}^d$, the affine certificate forces the operator identity $\sum_{a \in A} P_a = \sum_{b \in B} P_b$ in every FOR of H. That operator identity imposes geometric rigidity [2, 5]:

1. *No single complete nonorthogonal d-tuple:* If $U = \{u_1, \ldots, u_d\}$ satisfies $\sum_{u \in U} P_u = I_d$ in $\mathbb{C}^d$, then $P_{u_i} P_{u_j} = 0$ for $i \neq j$, so U must be an orthogonal basis rather than a nonorthogonal set.

2. *No size-one ($k = 1$) Hilbert-space pseudocontext:* $P_a = P_b$ implies $[q_a] = [q_b]$, violating ray injectivity.

3. *No size-two ($k = 2$) real Hilbert-space pseudocontext:* In $\mathbb{R}^d$, if $q_{a_1} \not\perp q_{a_2}$, the spectral decomposition of $P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2}$ uniquely fixes the unordered pair of rays $\{[q_{a_1}], [q_{a_2}]\} = \{[q_{b_1}], [q_{b_2}]\}$. Thus paired affine pseudocontexts require $k \geq 2$ over $\mathbb{C}$ (attained by a 20-ray octagon H_{20} in $\mathbb{C}^3$) and $k \geq 3$ over $\mathbb{R}$ (attained by Mayet's 15-ray hypergraph $M_{15,8}$ in $\mathbb{R}^3$) [2].

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A parallel line of research studies *common-spectrum spectral assignments* (strong d-colorings) of quantum hypergraphs [6–9]. Assigning each ray $v \in V$ a context-independent eigenvalue $\lambda_{c(v)}$ from a fixed d-element spectrum $\Sigma = \{\lambda_0, \ldots, \lambda_{d-1}\}$ defines nondegenerate commuting observables $A_e = \sum_{v \in e} \lambda_{c(v)} P_v$ [10, 11] if and only if $c : V \to \{0, \ldots, d-1\}$ is a strong d-coloring ($c \in \mathcal{C}_d(H)$), injective on every context $e \in E$. In the companion work [9], strong d-colorings were characterized by $n \times d$ *Context–Color Matrices* $M \in V^{n \times d}$ whose rows permute the d rays of each context and whose d columns are the *extendable* two-valued states $\mathcal{S}_{\text{ext}}(H) \subseteq \mathcal{S}(H)$.

The central question of this paper is different: can a set of pairwise nonadjacent vertices be forced to be complete and exclusive by all global strong d-colorings? We call such a set a *chromatic admissible context*. Its vertices occupy all d columns of every Context–Color Matrix, exactly once each. More generally, an *admissibility context relative to a nonempty state family* is an independent set whose state sum is identically one on that family. This formulation separates the chromatic case, based on $\mathcal{S}_{\text{ext}}(H)$, from the stronger two-valued case, based on all of $\mathcal{S}(H)$. Neither formulation presupposes an affine or operator identity. Alongside this central notion, we study universal rainbow exclusivity, chromatic twins, and paired color-multiset equivalence. We show that:

- A faithful real three-dimensional construction $H_{53,34}$ has two independent triples normalized on all 224 two-valued states. Pasting two nonseparating gadgets gives a short structural proof; an explicit Born preparation with total 8/9 separates two-valued state admissibility from affine and operator normalization (Theorem 8).
- Every balanced affine pseudocontext (A, B) automatically enforces *color-multiset conservation* $\{\{c(a) : a \in A\}\} = \{\{c(b) : b \in B\}\}$ in every $n \times d$ Context–Color Matrix (Theorem 3 and Corollary 4).
- Faithfully coordinatized real three-dimensional hypergraphs admit both *chromatic admissible contexts* (normalized on $\mathcal{S}_{\text{ext}}(H)$, starting at $H_{20,12}$ and $H_{21,13}$) and genuine *two-valued state admissibility contexts* (normalized on all of $\mathcal{S}(H)$, realized in $G_{41,25}$ and $G_{41,26}$ via cascaded roof-twin gadgets). The examples also exhibit non-twin color-multiset-equivalent pairs of size two, universally rainbow nonorthogonal subsets, and chromatic twin classes, with related phenomena in $\mathbb{R}^4$. Because $\{0,1\}$-integrality on Specker bugs enforces biconditional two-valued twins $s(x) = s(x')$ that do not hold on $\mathcal{A}(H)$ ($\text{aff}\,\mathcal{S}(H) \subsetneq \mathcal{A}(H)$), the operator rigidity statements above remain valid.

Section II defines admissibility relative to a state family and the related chromatic relations. Section III analyzes affine pseudocontexts and their color-multiset consequences. Section IV constructs a two-valued state admissibility context in $\mathbb{R}^3$, with its structural proof, full hypergraph drawing, and exact FOR. Section V then treats chromatic admissible contexts and color-multiset-equivalent pairs, including structural color propagation in the Yu–Oh example. Section V F develops the roof-twin cascades $G_{41,25}, G_{41,26}$ and binary state restrictions in the Yu–Oh examples. Section VI treats four-dimensional relations. Section VII compares the full and extendable classical envelopes with the quantum spectra for every configuration, and Section VIII gives the census and conclusion.

Section II A gives a general color-weighted projector construction and an exact maximum-expectation equality in at most d real color parameters; the final subsections of the example sections evaluate its eigensystems for the stated FORs.

II. CONTEXT–COLOR MATRICES AND ADMISSIBILITY CONTEXTS

Let $H = (V, E)$ be a finite d-uniform hypergraph with $V = \bigcup_{e \in E} e$ and $n = |E|$ contexts $E = \{C_1, \ldots, C_n\}$. Its two-section graph $\Gamma(H)$ has vertex set V and edges $\{u, v\}$ whenever $u \neq v$ share at least one context $C_i \in E$. A *faithful orthogonal representation* (FOR) in $\mathbb{F}^d$ assigns nonzero vectors $q_v \in \mathbb{F}^d$ to $v \in V$ such that $[q_u] \neq [q_v]$ for $u \neq v$ and $\langle q_u, q_v \rangle = 0$ if and only if $\{u, v\} \in E(\Gamma(H))$. Thus an independent set $U \subseteq V$ in $\Gamma(H)$ is a family of *pairwise nonorthogonal* rays.

By the Universal $n \times d$ Context–Color Matrix Criterion [9, Theorem 1], H admits a strong d-coloring $c \in \mathcal{C}_d(H)$ if and only if the vertices of each context C_i can be ordered into an $n \times d$ matrix $M = (M_{i,j}) \in V^{n \times d}$ ($1 \leq i \leq n$, $0 \leq j \leq d-1$) such that (1) row i is a permutation of C_i, and (2) each vertex $v \in V$ appears in a single column $j = c(v)$. The column vertex sets $S_j = \{M_{i,j} : 1 \leq i \leq n\} = c^{-1}(\{j\})$ form d pairwise disjoint two-valued state supports whose indicator functions $s_j = \mathbf{1}_{S_j} \in \mathcal{S}_{\text{ext}}(H)$ partition V:

$$\sum_{j=0}^{d-1} s_j(v) = 1 \qquad (\forall\, v \in V). \tag{5}$$

Throughout, N_c counts normalized strong colorings obtained by fixing the color order on one reference context. Every labeled strong d-coloring has a unique such normalization and is recovered by one of the $d!$ color permutations. All universal color relations below are invariant under those permutations, so normalized representatives cover every strong d-coloring of the prescribed palette.

In every Context–Color Matrix, each actual context is a complete rainbow row. The following definition isolates the same exclusivity and completeness when they are enforced on vertices that share no context with one another.

Definition 1 (Admissibility Context Relative to a State Family)**.** Let $\varnothing \neq \mathcal{T} \subseteq \mathcal{S}(H)$. An independent set $U \subseteq V$ in $\Gamma(H)$ is an *admissibility context relative to* $\mathcal{T}$ if

$$\sum_{u\in U} s(u) = 1 \qquad (\forall\, s \in \mathcal{T}). \tag{6}$$

The equation enforces both exclusivity (at most one vertex is assigned one) and completeness (at least one is assigned one), despite pairwise nonadjacency. For $\mathcal{T} = \mathcal{S}_{\text{ext}}(H) \neq \varnothing$, we call U a *chromatic admissible context*; for $\mathcal{T} = \mathcal{S}(H) \neq \varnothing$, we call it a *two-valued state admissibility context*. The state family is part of the definition. No affine certificate or projector normalization is assumed.

When $\mathcal{C}_d(H) \neq \varnothing$, every two-valued state admissibility context is a chromatic admissible context because $\mathcal{S}_{\text{ext}}(H) \subseteq \mathcal{S}(H)$. The converse need not hold. The chromatic case has exactly d vertices: for any coloring, sum Eq. (6) over its d color-class states and use Eq. (5) to obtain $|U| = d$. Theorem 3(1) identifies this condition with universal rainbow completeness.

We now distinguish this normalization property from the other universal chromatic relations used below.

Definition 2 (Admissibility and Related Chromatic Relations)**.** Let $H = (V, E)$ be a d-uniform hypergraph with $\mathcal{C}_d(H) \neq \varnothing$.

1. *Class I: Chromatic Admissible Contexts.* An independent set $U = \{u_1, \ldots, u_d\} \subseteq V$ is a chromatic admissible context precisely when it receives all d colors in every strong d-coloring:

$$\big\{c(u_1), \ldots, c(u_d)\big\} = \{0, 1, \ldots, d-1\} \qquad \forall\, c \in \mathcal{C}_d(H). \tag{7}$$

   If a non-context d-tuple satisfies Eq. (7) but has both adjacent and nonadjacent pairs, we call it a *partially nonorthogonal complete rainbow d-tuple*. It is not an admissibility context under Definition 1, which requires pairwise nonadjacency.

2. *Class II: Universal Rainbow Sets and Chromatic Twin Classes.* Let $U = \{u_1, \ldots, u_k\} \subseteq V$ be an independent set with $k \geq 2$.

   (a) U is a *universally rainbow (chromatically exclusive) k-tuple* ($2 \leq k < d$) if its elements receive pairwise distinct colors in every strong d-coloring:

$$c(u_a) \neq c(u_b) \quad (1 \leq a < b \leq k), \qquad \forall\, c \in \mathcal{C}_d(H). \tag{8}$$

   (b) Write $u \sim_c v$ if $c(u) = c(v)$ for every $c \in \mathcal{C}_d(H)$; such vertices are *chromatic twins*. A *chromatic twin class* is an equivalence class of $\sim_c$, and a *universally monochromatic set* is any subset of such a class. Thus U is universally monochromatic (with no upper bound on k) if

$$c(u_1) = c(u_2) = \cdots = c(u_k) \qquad \forall\, c \in \mathcal{C}_d(H). \tag{9}$$

3. *Class III: Color-Multiset-Equivalent Pairs.* Two disjoint independent sets $A = \{a_1, \ldots, a_k\}$ and $B = \{b_1, \ldots, b_k\}$, with $2 \leq k \leq d$, form a *color-multiset-equivalent pair* if

$$\big\{\{c(a) : a \in A\}\big\} = \big\{\{c(b) : b \in B\}\big\} \qquad \forall\, c \in \mathcal{C}_d(H). \tag{10}$$

   The pair is *non-twin* if no element of A is a chromatic twin of an element of B ($\neg(a_i \sim_c b_j)$ for all $a_i \in A$, $b_j \in B$). Color-multiset equivalence alone does not impose exclusivity or completeness on either side. We reserve *affine pseudocontext* for a pair with a balanced context certificate. In the census, we count non-twin pairs and omit pairs in which both sides are already chromatic admissible contexts; their equality follows directly from the single-set normalizations.

Theorem 3 (Chromatic Relations via Extendable State Sums)**.** *Let $H = (V, E)$ be a d-uniform hypergraph with $\mathcal{C}_d(H) \neq \varnothing$ and extendable state set $\mathcal{S}_{\text{ext}}(H) = \{s_j = \mathbf{1}_{c^{-1}(\{j\})} : c \in \mathcal{C}_d(H),\, 0 \leq j \leq d-1\}$. Then:*

1. *A d-element set $U = \{u_1, \ldots, u_d\} \subseteq V$ satisfies $\{c(u) : u \in U\} = \{0, \ldots, d-1\}$ for all $c \in \mathcal{C}_d(H)$ if and only if*

$$\sum_{u\in U} s(u) = 1 \qquad \forall\, s \in \mathcal{S}_{\text{ext}}(H). \tag{11}$$

2. *A k-element set $U = \{u_1, \ldots, u_k\} \subseteq V$ ($2 \leq k < d$) satisfies $c(u_a) \neq c(u_b)$ ($a \neq b$) for all $c \in \mathcal{C}_d(H)$ if and only if*

$$\sum_{u\in U} s(u) \leq 1 \qquad \forall\, s \in \mathcal{S}_{\text{ext}}(H). \tag{12}$$

3. *Disjoint k-element sets $A, B \subseteq V$ satisfy $\{\{c(a) : a \in A\}\} = \{\{c(b) : b \in B\}\}$ for all $c \in \mathcal{C}_d(H)$ if and only if*

$$\sum_{a\in A} s(a) = \sum_{b\in B} s(b) \qquad \forall\, s \in \mathcal{S}_{\text{ext}}(H). \tag{13}$$

Proof. For any $c \in \mathcal{C}_d(H)$, let $s_j = \mathbf{1}_{c^{-1}(\{j\})} \in \mathcal{S}_{\text{ext}}(H)$ ($0 \leq j \leq d-1$) be the indicator functions of the d columns of the associated $n \times d$ Context–Color Matrix M. For any subset $X \subseteq V$ and color $j \in \{0, \ldots, d-1\}$, the column sum

$$\sum_{x\in X} s_j(x) = \big|\{x \in X : c(x) = j\}\big| \tag{14}$$

counts the exact multiplicity of color j on X in coloring c. (1) When $|U| = d$, every color $j \in \{0, \ldots, d-1\}$ has multiplicity 1 on U iff $\{c(u) : u \in U\} = \{0, \ldots, d-1\}$, establishing Eq. (11). (2) Every color j has multiplicity at most 1 on U iff no two elements of U share a color in c, establishing Eq. (12). (3) By Eq. (14), color j has the same multiplicity on A and on B for every $j \in \{0, \ldots, d-1\}$ and every $c \in \mathcal{C}_d(H)$ iff $\sum_{a\in A} s_j(a) = \sum_{b\in B} s_j(b)$ for every $s_j \in \mathcal{S}_{\text{ext}}(H)$, establishing Eq. (13). □

Corollary 4 (Affine Pseudocontexts Imply Color-Multiset Equivalence)**.** *Let $M_H \in \{0,1\}^{n\times|V|}$ be the context-vertex incidence matrix of H. If disjoint nonorthogonal k-tuples $A, B \subseteq V$ admit a balanced context certificate $\lambda \in \mathbb{R}^n$ satisfying [2]*

$$M_H^{\mathsf{T}}\lambda = \chi_A - \chi_B, \qquad \mathbf{1}_n^{\mathsf{T}}\lambda = 0, \tag{15}$$

then in every strong d-coloring $c \in \mathcal{C}_d(H)$, A and B carry the exact same multiset of colors: $\{\{c(a) : a \in A\}\} = \{\{c(b) : b \in B\}\}$.

Proof. Every extendable state $s \in \mathcal{S}_{\text{ext}}(H) \subseteq \mathcal{S}(H)$ satisfies $M_H s = \mathbf{1}_n$, so

$$\sum_{a\in A} s(a) - \sum_{b\in B} s(b) = (\chi_A - \chi_B)^{\mathsf{T}} s = \lambda^{\mathsf{T}} M_H s = \lambda^{\mathsf{T}}\mathbf{1}_n = 0.$$

Applying Theorem 3(3) completes the proof. □

A. Color-Weighted Operators and Exact Quantum Maxima

Fix a FOR, write $\hat{q}_v = q_v/\|q_v\|$, and fix a strong coloring $\kappa \in \mathcal{C}_d(H)$. Assign the real number c_j to color j, with $\mathbf{c} = (c_0, \dots, c_{d-1}) \in \mathbb{R}^d$. Each actual context $C \in E$ then supplies the spectral sum

$$A_C(\mathbf{c};\kappa) = \sum_{v\in C} c_{\kappa(v)} P_v, \qquad P_v = \hat{q}_v \hat{q}_v^{\dagger}. \tag{16}$$

Its eigenvalues are precisely the assigned color values; distinct values give a nondegenerate, maximal context observable. For a pseudocontext or another target set $X \subseteq V$, form instead

$$O_X(\mathbf{c};\kappa) = \sum_{v\in X} c_{\kappa(v)} P_v = \sum_{j=0}^{d-1} c_j S_{X,j}, \quad S_{X,j} = \sum_{\substack{v\in X \\ \kappa(v)=j}} P_v. \tag{17}$$

This sum is Hermitian for arbitrary real color values. Since the target rays need not be orthogonal, Eq. (17) is generally not yet a spectral decomposition: the $S_{X,j}$ need not be projections, and the c_j need not be its eigenvalues. Its eigensystem supplies the spectral decomposition and the optimal quantum preparations, by the same min–max construction used for Bell operators in Ref. [12].

Proposition 5 (Parameterized Quantum Maximum)**.** *For fixed κ, fixed FOR, and fixed real $\mathbf{c}$, let Λ_X be the set of distinct eigenvalues of O_X. Then*

$$O_X = \sum_{\lambda\in\Lambda_X} \lambda \Pi_\lambda, \qquad \Pi_\lambda = \prod_{\substack{\mu\in\Lambda_X \\ \mu\neq\lambda}} \frac{O_X - \mu I_d}{\lambda - \mu}. \tag{18}$$

In particular, with $p_\rho(v) = \operatorname{Tr}(\rho P_v)$ and the maximum taken over $\rho \succeq 0$, $\operatorname{Tr}\rho = 1$,

$$\boxed{\begin{aligned} &\max_\rho \sum_{v\in X} c_{\kappa(v)} p_\rho(v) = \lambda_{\max}(O_X) \\ &= \max\{\lambda \in \mathbb{R} : \det(\lambda I_d - O_X) = 0\}. \end{aligned}} \tag{19}$$

Equality is attained exactly by density operators supported on $\operatorname{ran}\Pi_{\lambda_{\max}}$. The analogous minimum is $\lambda_{\min}(O_X)$, and the largest absolute expectation is $\|O_X\| = \max\{|\lambda_{\min}|, |\lambda_{\max}|\}$.

Proof. The spectral theorem gives Eq. (18); its polynomial expression projects onto the whole eigenspace even when an eigenvalue is degenerate. The expectation $\operatorname{Tr}(\rho O_X) = \sum_\lambda \lambda \operatorname{Tr}(\rho\Pi_\lambda)$ is a convex combination of eigenvalues. Its maximum is attained exactly when the total weight lies in the largest eigenspace. □

There is a finite polynomial equality for every such target in every dimension. Order $X = (x_1, \dots, x_k)$, put $Q = (\hat{q}_{x_1}, \dots, \hat{q}_{x_k})$, $G = Q^\dagger Q$, $a_i = c_{\kappa(x_i)}$, and $D = \operatorname{diag}(a_1, \dots, a_k)$. Thus $O_X = QDQ^\dagger$. The characteristic polynomial is

$$\begin{aligned} \det(\lambda I_d - O_X) = &\sum_{\ell=0}^{\min(k,d)} (-1)^\ell \lambda^{d-\ell} \\ &\times \sum_{\substack{J\subseteq\{1,\dots,k\} \\ |J|=\ell}} \det G[J] \prod_{i\in J} a_i, \end{aligned} \tag{20}$$

where the empty minor and product equal one. This follows by expanding the principal minors of $QDQ^\dagger$ and applying Cauchy–Binet. If $k \le d$, it is equivalently $\lambda^{d-k}\det(\lambda I_k - DG)$. For a nonzero root, $DGz = \lambda z$ gives the normalized eigenvector $\psi = Qz/\|Qz\|$; the zero eigenspace is $\ker O_X$. Alternatively, Eq. (18) gives every eigenspace directly, including all degeneracies and zero color values.

Only $e_X = |\kappa(X)| \le d$ color values occur in these equalities; after relabeling the used colors these are $c_0, \dots, c_{e_X-1}$. A rainbow d-tuple uses all d parameters. The representation and coloring are part of the data: the incidence structure alone does not fix the Gram minors. Maximizing over FORs is a further optimization. Likewise, varying unrestricted color values generally gives no finite overall maximum, since a common positive rescaling rescales the operator.

The relevant classical comparison must also specify its state family. For the d column states s_j of the *fixed* coloring κ,

$$\sum_{v\in X} c_{\kappa(v)} s_j(v) = c_j\, m_{X,j}, \qquad m_{X,j} = |X \cap \kappa^{-1}(\{j\})|. \tag{21}$$

For a d-tuple rainbow in this coloring, $m_{X,j} = 1$, so convex mixtures of these column states give expectations between $\min_j c_j$ and $\max_j c_j$. If X is a chromatic admissible context, every extendable state selects exactly one vertex of X, so this bound holds on all of $\operatorname{conv}\mathcal{S}_{\text{ext}}(H)$, including states generated by other colorings. For a two-valued state admissibility context, the bound holds on all of $\operatorname{conv}\mathcal{S}(H)$. Thus a quantum maximum above $\max_j c_j$, or a minimum below $\min_j c_j$, separates the corresponding state family. For a color-multiset-equivalent pair, $m_{A,j} = m_{B,j}$ gives

zero weighted difference on the column states of the fixed κ; this zero identity need not hold on states generated by another coloring. An all-state classical bound for a weighted pair must therefore be established separately. Setting every $c_j = 1$ recovers the unweighted state-sum statements proved above.

For the three-dimensional examples, the entire calculation reduces to a cubic. For an ordered triple (x_0, x_1, x_2) with weights $a_i = c_{\kappa(x_i)}$, define

$$\begin{aligned} h_{ij} &= 1 - |\langle \hat{q}_{x_i}, \hat{q}_{x_j} \rangle|^2, \qquad g = \det\big(\langle \hat{q}_{x_i}, \hat{q}_{x_j} \rangle\big)_{i,j=0}^{2}, \\ t &= a_0 + a_1 + a_2, \\ b &= h_{01}a_0a_1 + h_{02}a_0a_2 + h_{12}a_1a_2, \\ \delta &= g\, a_0a_1a_2. \end{aligned} \tag{22}$$

The eigenvalues are exactly the three real roots, with multiplicity, of

$$\lambda^3 - t\lambda^2 + b\lambda - \delta = 0. \tag{23}$$

To give an explicit expression for the maximum, put $\Delta = t^2 - 3b$. For $\Delta > 0$, define

$$\begin{aligned} \theta &= \frac{1}{3}\arccos\left(\frac{2t^3 - 9tb + 27\delta}{2\Delta^{3/2}}\right), \\ F(t, b, \delta) &= \frac{t + 2\sqrt{\Delta}\cos\theta}{3}. \end{aligned} \tag{24}$$

The other two eigenvalues replace θ by $\theta - 2\pi/3$ and $\theta - 4\pi/3$. If $\Delta = 0$, all three eigenvalues equal $t/3$, and we set $F = t/3$. Hermiticity ensures $\Delta \geq 0$ and an arccosine argument in $[-1, 1]$. Consequently, Eq. (19) is the explicit equality $\max_\rho \mathrm{Tr}(\rho O_X) = F(t, b, \delta)$ for every triple below, including repeated or negative color values.

B. Exact Classical Envelopes and the Quantum Comparison

The spectral maximum must be compared with the same weighted expression on a specified classical state family. Let $\varnothing \neq \mathcal{T} \subseteq \mathcal{S}(H)$, and write $f = (f_v)_{v \in V}$ for a signed target: $f = \chi_X$ for a single set, $f = \chi_A - \chi_B$ for a paired relation, or $f = \chi_{\{u\}} - \chi_{\{v\}}$ for a twin difference. For a fixed coloring κ, define the integer coefficient vector of a state by

$$n_j(s; f, \kappa) = \sum_{\substack{v \in V \\ \kappa(v) = j}} f_v s(v), \qquad 0 \leq j < d. \tag{25}$$

Discard duplicate vectors and put

$$K_{f,\kappa}(\mathcal{T}) = \mathrm{conv}\{n(s; f, \kappa) : s \in \mathcal{T}\}. \tag{26}$$

State multiplicities do not affect this polytope. In particular, there is no assumption that the classical preparation averages uniformly over the enumerated states.

Proposition 6 (Classical Bounds and Spectral Separation)**.** *For arbitrary real color values, the sharp classical expectation interval over* $\mathrm{conv}\,\mathcal{T}$ *has endpoints*

$$\begin{aligned} \beta_{\mathcal{T}}^{-}(\mathbf{c}) &= \min_{s \in \mathcal{T}} \sum_j c_j n_j(s; f, \kappa), \\ \beta_{\mathcal{T}}^{+}(\mathbf{c}) &= \max_{s \in \mathcal{T}} \sum_j c_j n_j(s; f, \kappa). \end{aligned} \tag{27}$$

For the same FOR and coloring, construct

$$O_f(\mathbf{c}; \kappa) = \sum_v f_v c_{\kappa(v)} P_v. \tag{28}$$

Its sharp quantum expectation interval is $[\lambda_{\min}(O_f), \lambda_{\max}(O_f)]$. *A separation from* $\mathrm{conv}\,\mathcal{T}$ *occurs exactly when at least one of*

$$\begin{aligned} g_+(\mathbf{c}) &= \lambda_{\max}(O_f) - \beta_{\mathcal{T}}^{+}(\mathbf{c}), \\ g_-(\mathbf{c}) &= \beta_{\mathcal{T}}^{-}(\mathbf{c}) - \lambda_{\min}(O_f) \end{aligned} \tag{29}$$

is positive. The corresponding extremal eigenstate attains that separation.

Proof. A classical mixture has expectation $\sum_s \mu_s\, \mathbf{c} \cdot n(s)$, where $\mu_s \geq 0$ and $\sum_s \mu_s = 1$. Its extrema are attained at individual states, proving Eq. (27). The quantum endpoints follow from Proposition 5. The lower-bound test is equivalently the upper-bound test for $-O_f$. □

Thus the classical endpoints are exact piecewise-linear functions of at most d parameters, while the quantum endpoints are the smallest and largest roots of the characteristic equality. Both endpoints matter. If U is an admissibility context on either full family $\mathcal{S}(H)$ or $\mathcal{S}_{\mathrm{ext}}(H)$ considered here, its distinct restrictions are all the one-hot vectors: every state selects exactly one target vertex, and the fixed coloring's column states realize every choice. Hence

$$\beta_{\mathcal{T}}^{-}(\mathbf{c}) = \min_j c_j, \qquad \beta_{\mathcal{T}}^{+}(\mathbf{c}) = \max_j c_j. \tag{30}$$

For a general smaller state family, the minimum and maximum are taken only over those one-hot restrictions that actually occur. Equation (30) holds on all of $\mathcal{S}(H)$ only for two-valued state admissibility; chromatic admissibility initially gives it on $\mathcal{S}_{\mathrm{ext}}(H)$. For a weighted paired relation, the zero difference on the columns of κ need not be its bound on either complete state family.

A finite parameter test is available without sampling color values. Write the exact polytope $K_{f,\kappa}(\mathcal{T})$ as affine equalities and facet inequalities. For each facet $a \cdot n \leq b$, compare $\lambda_{\max}(O_f(a; \kappa))$ with b; test both signs of each affine equality. These tests determine whether the quantum expectation image is contained in the classical polytope, and hence whether any real color weights can separate it. They test the existence of a separating direction, not the largest possible gap under a chosen coefficient normalization. To optimize gap magnitude, a normalization such as $\|\mathbf{c}\|_2 = 1$ must first be imposed.

The accompanying calculation applies this construction to both $\mathcal{T} = \mathcal{S}(H)$ and $\mathcal{T} = \mathcal{S}_{\text{ext}}(H)$ for all 19 configurations. Tkadlec's block-recursion program [13] generates the two-valued states; an independently implemented propagation solver checks the complete state lists. Section VII gives the exhaustive comparison and its verification details.

III. PAIRED AFFINE-FORCED PSEUDOCONTEXTS IN $\mathbb{R}^3$ AND $\mathbb{C}^3$

A. Mayet's 15-Ray, 8-Context Real Configuration $M_{15,8}$

Consider the 15-vertex, 8-context hypergraph $M_{15,8}$ studied in Refs. [1, 2, 14, 15], with contexts

$$\begin{aligned} C_1 &= \{1,2,3\}, & C_2 &= \{3,4,5\}, \\ C_3 &= \{6,7,8\}, & C_4 &= \{8,9,10\}, \\ C_5 &= \{11,12,13\}, & C_6 &= \{13,14,15\}, \\ C_7 &= \{2,7,12\}, & C_8 &= \{4,9,14\}. \end{aligned} \tag{31}$$

Clearing denominators in Eqs. (9) of Ref. [1] supplies an exact integer FOR in $\mathbb{R}^3$ (zero extra orthogonalities among all $\binom{15}{2} = 105$ pairs):

$$\begin{aligned} r_1 &= (5,3,-4), & r_2 &= (5,-7,1), \\ r_3 &= (1,1,2), & r_4 &= (1,1,-1), \\ r_5 &= (1,-1,0), & r_6 &= (5,-16,13), \\ r_7 &= (5,4,3), & r_8 &= (-2,1,2), \\ r_9 &= (1,0,1), & r_{10} &= (1,4,-1), \\ r_{11} &= (-5,18,1), & r_{12} &= (5,2,-11), \\ r_{13} &= (4,1,2), & r_{14} &= (-1,2,1), \\ r_{15} &= (1,2,-3). \end{aligned} \tag{32}$$

The two primary target triples $A = \{1,6,11\}$ and $B = \{5,10,15\}$ are internally pairwise nonorthogonal. The two 4-context coverings $\mathcal{F}_+ = \{C_1, C_3, C_5, C_8\}$ and $\mathcal{F}_- = \{C_2, C_4, C_6, C_7\}$ partition $V \setminus B$ and $V \setminus A$, respectively, giving $\sum_{C\in\mathcal{F}_+} \chi_C - \sum_{C\in\mathcal{F}_-} \chi_C = \chi_A - \chi_B$. In addition, omitting any one of the three horizontal 2-context chains (C_1, C_2), (C_3, C_4), or (C_5, C_6) leaves a 6-context hexagon whose 3-context coverings force three further paired size-three affine pseudocontexts: $\{1,6,14\}$ vs. $\{5,10,12\}$, $\{1,9,11\}$ vs. $\{5,7,15\}$, and $\{4,6,11\}$ vs. $\{2,10,15\}$ (4 pairs in total).

Exhaustive enumeration gives 24 two-valued states of $M_{15,8}$, all extendable, and $N_c = 16$ normalized 8×3 Context–Color Matrices (with anchor $C_1 = (1,2,3)$). By Corollary 4, the two target triples have identical color multisets in every coloring:

- In 4 normalized colorings, both A and B are *rainbow* $\{0,1,2\}$.
- In 12 normalized colorings, both A and B carry a *two-color multiset*: 4 with $\{0,0,1\}$, 4 with $\{0,1,1\}$, and 4 with $\{0,0,2\}$. (Neither triple is ever monochromatic because $p(1)+p(6)+p(11) \le 2$ on $\mathcal{S}(M_{15,8})$ [1].)

Two explicit 8×3 Context–Color Matrices illustrating the rainbow regime ($M_1^{(M_{15,8})}$) and the two-color multiset regime ($M_2^{(M_{15,8})}$) are:

$$M_1^{(M_{15,8})} = \left(\begin{array}{ccc} \mathbf{c}_0 & \mathbf{c}_1 & \mathbf{c}_2 \\ \hline \mathbf{1} & 2 & 3 \\ 4 & \mathbf{5} & 3 \\ 8 & \mathbf{6} & 7 \\ 8 & 9 & \mathbf{10} \\ 12 & 13 & \mathbf{11} \\ \mathbf{15} & 13 & 14 \\ \hline 12 & 2 & 7 \\ 4 & 9 & 14 \end{array}\right), \quad M_2^{(M_{15,8})} = \left(\begin{array}{ccc} \mathbf{c}_0 & \mathbf{c}_1 & \mathbf{c}_2 \\ \hline \mathbf{1} & 2 & 3 \\ 4 & \mathbf{5} & 3 \\ \mathbf{6} & 8 & 7 \\ \mathbf{10} & 8 & 9 \\ 12 & \mathbf{11} & 13 \\ \mathbf{15} & 14 & 13 \\ \hline 12 & 2 & 7 \\ 4 & 14 & 9 \end{array}\right). \tag{33}$$

In $M_1^{(M_{15,8})}$, both $A = \{1,6,11\}$ and $B = \{5,10,15\}$ span all three columns $(\mathbf{c}_0, \mathbf{c}_1, \mathbf{c}_2)$ via $(1,6,11)$ and $(15,5,10)$. In $M_2^{(M_{15,8})}$, both A and B have two entries in Column $\mathbf{c}_0$ ($\{1,6\}$ and $\{10,15\}$), one entry in Column $\mathbf{c}_1$ (11 and 5), and none in Column $\mathbf{c}_2$. Figure 1 displays both colorings on the Greechie diagram of $M_{15,8}$.

B. Balanced Size-Three Pseudocontexts in $B_{13,7}$ and $R_{21,12}, R_{25,15}$

Two further families of faithfully coordinatized hypergraphs from Ref. [9] already contain balanced covering certificates for size-three affine pseudocontexts:

1. *The 13-ray, 7-context Specker bug $B_{13,7} \subset \mathbb{R}^3$ ($N_c = 6$, $|\mathcal{S}_{\text{ext}}|/|\mathcal{S}| = 12/14$):* Its 7 contexts are $E_1 = \{1,2,3\}$, $E_2 = \{7,8,9\}$, $E_3 = \{2,8,5\}$, $E_4 = \{3,11,14\}$, $E_5 = \{7,11,15\}$, $E_6 = \{9,10,23\}$, and $E_7 = \{1,10,19\}$ [Fig. 4(a)], with Yu–Oh coordinates from Table II. The six outer contexts partition into two 3-context coverings $\mathcal{F}_+ = \{E_1, E_5, E_6\}$ and $\mathcal{F}_- = \{E_2, E_4, E_7\}$ of $(V \setminus \{5\}) \setminus B$ and $(V \setminus \{5\}) \setminus A$, where

$$A = \{2,15,23\}, \qquad B = \{8,14,19\}. \tag{34}$$

   Hence $(\chi_{E_1} + \chi_{E_5} + \chi_{E_6}) - (\chi_{E_2} + \chi_{E_4} + \chi_{E_7}) = \chi_A - \chi_B$, forcing $\{\{c(2), c(15), c(23)\}\} = \{\{c(8), c(14), c(19)\}\}$ across all $N_c = 6$ normalized 7×3 Context–Color Matrices (and likewise in the 4-dimensional common-apex lift $B_{14,7}^{(4)}$).

2. *Gaussian-integer row–column hub grids $R_{21,12}$ and $R_{25,15}$ in $\mathbb{C}^3$ [9, Sec. IV D]:* In $R_{21,12}$ ($N_c = 32$, $|\mathcal{S}_{\text{ext}}|/|\mathcal{S}| = 48/54$) and $R_{25,15}$ ($N_c = 24$, $|\mathcal{S}_{\text{ext}}|/|\mathcal{S}| = 48/81$), the 3×3 core grid has row contexts $R_1 = \{1,2,3\}$, $R_2 = \{4,5,6\}$, $R_3 = \{7,8,9\}$

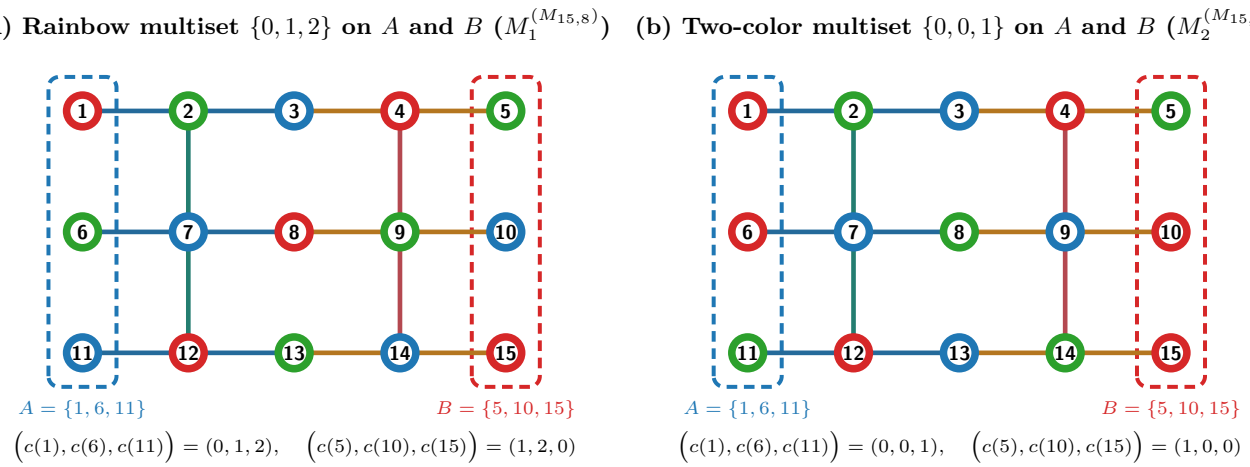


FIG. 1. Mayet's 15-ray, 8-context hypergraph $M_{15,8}$ in $\mathbb{R}^3$ [Eq. (32)] with paired size-three affine pseudocontexts $A = \{1,6,11\}$ and $B = \{5,10,15\}$ (dashed boxes). Vertex outer rings indicate colors 0 (red), 1 (green), and 2 (blue) for the two Context–Color Matrices in Eq. (33): (a) $M_1^{(M_{15,8})}$, where both A and B are rainbow $\{0,1,2\}$; (b) $M_2^{(M_{15,8})}$, where both A and B carry the two-color multiset $\{0,0,1\}$.

and hub contexts $\{1,10,13\}$, $\{2,11,14\}$, $\{3,12,15\}$ (for R_1), $\{4,10,16\}$, $\{5,11,17\}$, $\{6,12,18\}$ (for R_2), and $\{7,10,19\}$, $\{8,11,20\}$, $\{9,12,21\}$ (for R_3). For any two rows (say R_1 and R_2) and any column $j \in \{1,2,3\}$ (say $j = 1$), the two 3-context families $\mathcal{F}_+ = \{\{1,2,3\},\{4,10,16\},\{2,11,14\},\{3,12,15\}\} \setminus \{\{1,2,3\}\}$ combine to yield 12 size-three affine pseudocontexts such as

$$A = \{1,17,18\}, \qquad B = \{4,14,15\}, \tag{35}$$

via the balanced 3-context identity $(\chi_{\{1,2,3\}} + \chi_{\{5,11,17\}} + \chi_{\{6,12,18\}}) - (\chi_{\{4,5,6\}} + \chi_{\{2,11,14\}} + \chi_{\{3,12,15\}}) = \chi_{\{1,17,18\}} - \chi_{\{4,14,15\}}$, forcing $\{\{c(1),c(17),c(18)\}\} = \{\{c(4),c(14),c(15)\}\}$ in all 32 (resp. 24) normalized Context–Color Matrices.

C. The 20-Ray, 12-Context Complex Octagon $\mathsf{H}_{20} \subset \mathbb{C}^3$

To reach target size $k = 2$ via a balanced covering certificate, one must pass from $\mathbb{R}^3$ to $\mathbb{C}^3$ [2, 5]. The 20-ray, 12-context octagon H_{20} (Fig. 2) has eight perimeter contexts $E_j = \{v_{j-1,j}, v_j, v_{j,j+1}\}$ ($j \in \mathbb{Z}_8$) and four inner contexts $D_0 = \{v_0, a_1, v_4\}$, $D_2 = \{v_2, a_2, v_6\}$, $D_1 = \{v_1, b_2, v_5\}$, $D_3 = \{v_3, b_1, v_7\}$. The two 6-context coverings $\mathcal{G}_+ = \{D_0, D_2, E_1, E_3, E_5, E_7\}$ and $\mathcal{G}_- = \{D_1, D_3, E_0, E_2, E_4, E_6\}$ yield $\sum_{C\in\mathcal{G}_+} \chi_C - \sum_{C\in\mathcal{G}_-} \chi_C = \chi_A - \chi_B$ for the nonorthogonal pairs

$$A = \{a_1, a_2\}, \qquad B = \{b_1, b_2\}. \tag{36}$$

Exhaustive computation gives $|\mathcal{S}(\mathsf{H}_{20})| = |\mathcal{S}_{\text{ext}}(\mathsf{H}_{20})| = 26$ two-valued states and $N_c = 12$ normalized 12×3 Context–Color Matrices (with anchor $E_0 = (v_{70}, v_0, v_{01})$). Across all 12 normalized colorings, $\{\{c(a_1), c(a_2)\}\} = \{\{c(b_1), c(b_2)\}\}$ holds identically: 4 colorings assign monochromatic multisets (2 with $\{0,0\}$ and 2 with $\{2,2\}$) and 8 assign bichromatic multisets (2 with $\{0,1\}$, 4 with $\{0,2\}$, and 2 with $\{1,2\}$). Representative bichromatic ($M_1^{(\mathsf{H}_{20})}$) and monochromatic ($M_2^{(\mathsf{H}_{20})}$) Context–Color Matrices are:

$$M_1^{(\mathsf{H}_{20})} = \left(\begin{array}{ccc} \mathbf{c_0} & \mathbf{c_1} & \mathbf{c_2} \\ \hline v_{70} & v_0 & v_{01} \\ v_1 & v_{12} & v_{01} \\ v_{23} & v_{12} & v_2 \\ v_{23} & v_{34} & v_3 \\ v_4 & v_{34} & v_{45} \\ v_{56} & v_5 & v_{45} \\ v_{56} & v_6 & v_{67} \\ v_{70} & v_7 & v_{67} \\ \hline v_4 & v_0 & \mathbf{a_1} \\ \mathbf{a_2} & v_6 & v_2 \\ v_1 & v_5 & \mathbf{b_2} \\ \mathbf{b_1} & v_7 & v_3 \end{array}\right), \quad M_2^{(\mathsf{H}_{20})} = \left(\begin{array}{ccc} \mathbf{c_0} & \mathbf{c_1} & \mathbf{c_2} \\ \hline v_{70} & v_0 & v_{01} \\ v_{12} & v_1 & v_{01} \\ v_{12} & v_2 & v_{23} \\ v_{34} & v_3 & v_{23} \\ v_{34} & v_{45} & v_4 \\ v_{56} & v_{45} & v_5 \\ v_{56} & v_{67} & v_6 \\ v_{70} & v_{67} & v_7 \\ \hline \mathbf{a_1} & v_0 & v_4 \\ \mathbf{a_2} & v_2 & v_6 \\ \mathbf{b_2} & v_1 & v_5 \\ \mathbf{b_1} & v_3 & v_7 \end{array}\right). \tag{37}$$

D. Color-Weighted Eigensystems of the Affine Pairs

The affine certificate gives $O_A(\mathbf{1}) = O_B(\mathbf{1})$ in every FOR. For general color values, construct the signed operator

$$\begin{aligned} D_{A,B}(\mathbf{c};\kappa) &= O_A(\mathbf{c};\kappa) - O_B(\mathbf{c};\kappa), \\ \max_\rho \operatorname{Tr}(\rho D_{A,B}) &= \lambda_{\max}(D_{A,B}). \end{aligned} \tag{38}$$

Color-multiset conservation implies $\operatorname{Tr} D_{A,B} = 0$, whereas the affine certificate implies invariance under $\mathbf{c} \mapsto \mathbf{c} + \tau\mathbf{1}$.

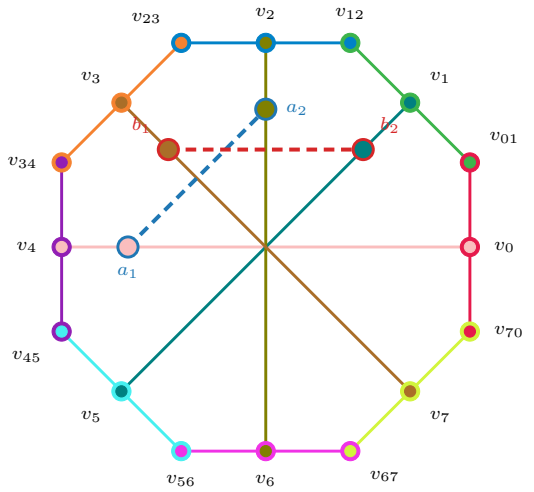


FIG. 2. The 20-ray, 12-context complex octagon $\mathsf{H}_{20} \subset \mathbb{C}^3$ [2, 5]. The two nonorthogonal target pairs $A = \{a_1, a_2\}$ (blue dashed link) and $B = \{b_1, b_2\}$ (red dashed link) carry identical 2-element color multisets $\{\{c(a_1), c(a_2)\}\} = \{\{c(b_1), c(b_2)\}\}$ across all $N_c = 12$ normalized 12×3 Context–Color Matrices [Eq. (37)].

It does not generally imply $D_{A,B} = 0$ for unequal color values. The maximum is positive whenever $D_{A,B} \neq 0$, because a nonzero traceless Hermitian operator has both positive and negative eigenvalues. In dimension three its characteristic equality is

$$\lambda^3 - \tfrac{1}{2}\operatorname{Tr}(D_{A,B}^2)\lambda - \det D_{A,B} = 0. \tag{39}$$

Its maximum is $F(0, -\operatorname{Tr}(D_{A,B}^2)/2, \det D_{A,B})$, with eigenspaces given by Eq. (18).

For Mayet's rainbow coloring $M_1^{(M_{15,8})}$, the ordered triples $A = (1, 6, 11)$ and $B = (5, 10, 15)$ carry weights (c_0, c_1, c_2) and (c_1, c_2, c_0), respectively. Directly forming the projectors from Eq. (32) gives

$$\begin{aligned} t &= c_0 + c_1 + c_2, \qquad \delta = \tfrac{2}{7} c_0 c_1 c_2, \\ b_A &= \tfrac{3}{4} c_0 c_1 + \tfrac{27}{28} c_0 c_2 + \tfrac{3}{7} c_1 c_2, \\ b_B &= \tfrac{3}{4} c_1 c_2 + \tfrac{27}{28} c_0 c_1 + \tfrac{3}{7} c_0 c_2, \\ \max_\rho \operatorname{Tr}(\rho O_A) &= F(t, b_A, \delta), \\ \max_\rho \operatorname{Tr}(\rho O_B) &= F(t, b_B, \delta). \end{aligned} \tag{40}$$

For their difference, the two coefficients in Eq. (39) are explicitly

$$\begin{aligned} \tfrac{1}{2}\operatorname{Tr} D_{A,B}^2 &= \tfrac{3}{700}\big(57c_0^2 + 119c_1^2 + 72c_2^2 \\ &\qquad - 104c_0c_1 - 10c_0c_2 - 134c_1c_2\big), \\ \det D_{A,B} &= \tfrac{27}{350}(c_0 - c_1)(c_0 - c_2)(c_1 - c_2). \end{aligned} \tag{41}$$

At $c_0 = c_1 = c_2 = 1$, the common operator has eigenvalues 2 and $\frac{1}{2} \pm \frac{\sqrt{21}}{14}$, so its maximal expectation is exactly 2. The two-color coloring $M_2^{(M_{15,8})}$ is covered by the same projector construction, with weights (c_0, c_0, c_1) on A and (c_1, c_0, c_0) on B.

TABLE I. Exact Gram coefficients for the ordered affine target triples. Both listed sides have the coefficients shown, in their respective displayed order. Color weights must be assigned according to the chosen coloring.

Ordered targets	h_{01}	h_{02}	h_{12}	g
$M_{15,8}$: $(1,6,11)$, $(5,10,15)$	3/4	27/28	3/7	2/7
$B_{13,7}$: $(2,15,23)$, $(8,14,19)$	5/6	5/6	5/9	4/9
$R_{21,12}, R_{25,15}$: $(1,17,18)$, $(4,14,15)$	9/10	9/10	3/4	13/20

The other displayed size-three affine pairs have equally short characteristic equalities. Table I gives the four coordinate-dependent coefficients of Eq. (22). The weights a_i are the color values on the ordered vertices, so repeated colors are included. In every row,

$$\max_\rho \operatorname{Tr}(\rho O_X) = F\left(\sum_i a_i, \sum_{i<j} h_{ij} a_i a_j, g \prod_i a_i\right). \tag{42}$$

The Gaussian-integer coordinates for the hub-grid row are those printed in Ref. [9, Sec. IV D]; the same target rays occur in $R_{21,12}$ and $R_{25,15}$. The principal-minor formula also treats the remaining affine pairs without an additional assumption about their color patterns.

For the complex octagon, choose the target coordinates of the FOR in Ref. [5],

$$\begin{aligned} q_{a_1} &= (s, 1, 0), & q_{a_2} &= (s, -1, 0), \\ q_{b_1} &= (s, \eta, 0), & q_{b_2} &= (s, -\eta, 0), \\ s^2 &= \frac{49 - 5\sqrt{69}}{26}, & \eta &= \frac{5 + 12i}{13}. \end{aligned} \tag{43}$$

Here $|\langle \hat{q}_{a_1}, \hat{q}_{a_2} \rangle|^2 = |\langle \hat{q}_{b_1}, \hat{q}_{b_2} \rangle|^2 = 23/75$. In the bichromatic coloring $M_1^{(\mathsf{H}_{20})}$, A has weights (c_2, c_0) and B has weights (c_0, c_2). Both weighted operators have eigenvalues

$$0, \qquad \lambda_\pm = \frac{c_0 + c_2 \pm \sqrt{(c_0 - c_2)^2 + \frac{92}{75} c_0 c_2}}{2}. \tag{44}$$

Their individual maximal expectations equal $\max\{0, \lambda_+\}$. They are isospectral even though their difference need not vanish. Indeed,

$$\begin{aligned} D_{A,B} &= (c_2 - c_0)(P_{a_1} - P_{b_2}), \\ \operatorname{spec} D_{A,B} &= \left\{0, \ \pm \tfrac{2\sqrt{3}}{5}(c_2 - c_0)\right\}, \\ \max_\rho \operatorname{Tr}(\rho D_{A,B}) &= \tfrac{2\sqrt{3}}{5}|c_2 - c_0|. \end{aligned} \tag{45}$$

For an explicit eigensystem, put $z = s(1 + \bar{\eta})/(1 + s^2)$, so $|z| = 2\sqrt{3}/5$. The unit vectors $2^{-1/2}(z/|z|, \pm 1, 0)^{\mathsf{T}}$ have eigenvalues $\pm(c_2 - c_0)|z|$, while $(0, 0, 1)^{\mathsf{T}}$ has eigenvalue zero. If $c_2 = c_0$, the whole space is the zero eigenspace. In the monochromatic coloring $M_2^{(\mathsf{H}_{20})}$, $D_{A,B} = 0$ for all color values, and the two individual spectra are $\{0, c_0(1 \pm \sqrt{23/75})\}$.

E. Classical Bounds for the Affine Pairs

For Mayet's ordered target $A = (1, 6, 11)$ in its rainbow coloring, all 24 two-valued states restrict to the seven binary patterns other than $(1, 1, 1)$. The exact classical envelope is therefore the minimum and maximum of

$$\{0, c_0, c_1, c_2, c_0 + c_1, c_0 + c_2, c_1 + c_2\}. \tag{46}$$

At unit values the classical interval is $[0, 2]$, containing the quantum interval $[\frac{1}{2} - \sqrt{21}/14, 2]$. For the signed pair in the displayed rainbow coloring, the distinct coefficient vectors are zero and $e_i - e_j$ $(i \neq j)$. Consequently,

$$\begin{aligned} \beta^+_{\mathcal{S}}(\mathbf{c}) &= \max_j c_j - \min_j c_j, \\ \beta^-_{\mathcal{S}}(\mathbf{c}) &= -\beta^+_{\mathcal{S}}(\mathbf{c}). \end{aligned} \tag{47}$$

This is the classical comparison for the spectrum in Eq. (41); positivity of its largest eigenvalue alone is not a violation of Eq. (47).

For the complex octagon's size-two target, all four binary restrictions occur, so the individual weighted bounds are

$$\beta^-_{\mathcal{S}} = \sum_{j \in \kappa(A)} \min(0, c_j), \qquad \beta^+_{\mathcal{S}} = \sum_{j \in \kappa(A)} \max(0, c_j) \tag{48}$$

in a bichromatic coloring. Its unit-weight quantum interval $[0, 1 + \sqrt{23/75}]$ is contained in the classical interval $[0, 2]$. The primary triples of $R_{21,12}$ and $R_{25,15}$ realize all eight binary patterns; their individual rainbow-weight bounds are similarly the sums of the negative and positive parts of the three color values. The full facet/equality comparison finds no separating color weights among the tested single or paired expressions for these two FORs, Mayet's FOR, or the complex octagon FOR. The exact envelopes of every color pattern and every pair are retained in the computational tables.

The enumeration also corrects the octagon census: besides its single size-two pair, it has 20 non-twin size-three pairs satisfying Definition 2. All 20 admit balanced affine certificates. For example,

$$\begin{aligned} A_3 &= \{a_1, v_{12}, v_{56}\}, \\ B_3 &= \{b_2, v_{34}, v_{70}\}, \\ \chi_{A_3} - \chi_{B_3} &= \chi_{D_0} + \chi_{E_1} + \chi_{E_5} \\ &\quad - \chi_{D_1} - \chi_{E_0} - \chi_{E_4}. \end{aligned} \tag{49}$$

Both triples are independent and disjoint in the same FOR. The original manuscript verifier independently reproduces the count 20, now recorded in Table IV.

IV. A TWO-VALUED STATE ADMISSIBILITY CONTEXT IN $\mathbb{R}^3$

We first realize the stronger alternative in Definition 1: an independent triple normalized on *every* two-valued state. The construction pastes two standard nonseparating gadgets at one ray. Its logical proof is independent of the subsequent enumeration, and an explicit FOR makes the distinction from affine and operator normalization concrete.

A. A Completed Nonseparating Gadget

Write $\mathcal{B}(a, b; x)$ for the seven-context Specker bug

$$\begin{gathered} \{a, x_1, x_2\}, \quad \{x_2, x_3, x_4\}, \\ \{x_4, x_5, b\}, \quad \{b, x_6, x_7\}, \\ \{x_7, x_8, x_9\}, \quad \{x_9, x_{10}, a\}, \\ \{x_3, x_8, x_{11}\}. \end{gathered} \tag{50}$$

Every two-valued state satisfies $s(a) + s(b) \leq 1$. Indeed, assigning one to both endpoints forces $s(x_2) = s(x_4) = s(x_7) = s(x_9) = 0$, hence $s(x_3) = s(x_8) = 1$, contradicting the final context.

Let G be the 27-ray, 17-context gadget formed by

$$\begin{gathered} \mathcal{B}(a, b; x), \quad \mathcal{B}(b', a'; y), \\ \{a, c, b'\}, \quad \{b, c, a'\}, \quad \{c, x_{11}, y_{11}\}. \end{gathered} \tag{51}$$

This is a completed realization of the Kochen–Specker Γ_3 nonseparating construction [16, 17]. The last context completes an orthogonal triple in the FOR below; the logical identity already follows from the other 16 contexts. In Fig. 3, the labels E_{15}, E_{16}, E_{17} follow the order of these three additional contexts; E_{32}, E_{33}, E_{34} label the corresponding contexts of the second copy. Each repeated label pairs the two branches of one context through its intertwining vertex.

Lemma 7 (Two-Valued Nonseparation)**.** *Every $s \in \mathcal{S}(G)$ satisfies*

$$\begin{gathered} s(a) = s(a'), \qquad s(b) = s(b'), \\ s(a) + s(b) + s(c) = 1. \end{gathered} \tag{52}$$

Proof. If $s(a) = 1$, the first bug forces $s(b) = 0$, and $\{a, c, b'\}$ forces $s(c) = s(b') = 0$. The context $\{b, c, a'\}$ then gives $s(a') = 1$. Conversely, if $s(a') = 1$, the second bug forces $s(b') = 0$, while $\{b, c, a'\}$ forces $s(b) = s(c) = 0$; now $\{a, c, b'\}$ gives $s(a) = 1$. Thus $s(a) = s(a')$. The symmetric argument gives $s(b) = s(b')$. Substitution into $s(a) + s(c) + s(b') = 1$ proves the final identity. □

B. One-Vertex Pasting and Universal Admissibility

Take a second copy $\widetilde{G}$ and identify only its $\widetilde{a}$ with the first copy's c. Write $u = \widetilde{a}'$. The union $H_{53,34} = G \cup \widetilde{G}$ has $27 + 27 - 1 = 53$ rays and $17 + 17 = 34$ contexts, all drawn in Fig. 3.

Theorem 8 (Two-Valued State Admissibility on Nonadjacent Rays)**.** *The hypergraph $H_{53,34}$ admits a FOR in $\mathbb{R}^3$ in which $U = \{a, b, u\}$ is pairwise nonorthogonal. Its two-valued state space is nonempty and*

$$s(a) + s(b) + s(u) = 1 \qquad \forall s \in \mathcal{S}(H_{53,34}). \tag{53}$$

Thus U is a two-valued state admissibility context and, in particular, a chromatic admissible context.

Proof. Restrict any two-valued state to each copy of G. Lemma 7 gives $s(a) + s(b) + s(c) = 1$ in the first copy and $s(c) = s(u)$ in the second, proving Eq. (53). The representation below verifies pairwise nonorthogonality and faithfulness. The enumeration in Eq. (61) establishes nonemptiness and shows that each distinguished vertex can be true. Every color-class state of every strong 3-coloring is a two-valued state, so Theorem 3(1) gives universal chromatic completeness as well. □

C. An Explicit Faithful Real Representation

Use the unnormalized ray representatives

$$\begin{aligned} q_a &= (1, -1, 1)^{\mathsf{T}}, \\ q_b &= (1, -1, -1)^{\mathsf{T}}, \\ q_c &= (1, 1, 0)^{\mathsf{T}}, \end{aligned} \tag{54}$$

and the bug vectors

i	$q_{x_i}^{\mathsf{T}}$	i	$q_{x_i}^{\mathsf{T}}$
1	$(2, 1, -1)$	7	$(1, 0, 1)$
2	$(0, 1, 1)$	8	$(0, 1, 0)$
3	$(1, 0, 0)$	9	$(1, 0, -1)$
4	$(0, 1, -1)$	10	$(1, 2, 1)$
5	$(2, 1, 1)$	11	$(0, 0, 1)$
6	$(1, 2, -1)$		

(55)

Put $n = q_c/\sqrt{2}$ and $R = nn^{\mathsf{T}} + [n]_\times$, where $[n]_\times v = n \times v$. This is a 90° rotation about n. The remaining rays of G are

$$q_{b'} = Rq_a, \quad q_{a'} = Rq_b, \quad q_{y_i} = Rq_{x_i}. \tag{56}$$

For the second copy, set $Q = L_0 E_0 F_0^{\mathsf{T}}$, with

$$\begin{aligned} F_0 &= \left(\frac{q_a}{\sqrt{3}} \;\; \frac{(1, -1, -2)^{\mathsf{T}}}{\sqrt{6}} \;\; n \right), \\ E_0 &= \left(n \;\; (0, 0, 1)^{\mathsf{T}} \;\; \frac{(1, -1, 0)^{\mathsf{T}}}{\sqrt{2}} \right), \\ L_0 &= \frac{1}{11} \begin{pmatrix} 9 & 2 & 6 \\ 2 & 9 & -6 \\ -6 & 6 & 7 \end{pmatrix}. \end{aligned} \tag{57}$$

The columns of E_0, F_0 are orthonormal, and L_0 is a rotation fixing n. Consequently, Qq_a is parallel to q_c, as required for the identified vertex. Assign Qq_v to every other ray of the second copy. In particular,

$$Qq_{a'} = -\frac{1}{11\sqrt{3}}(16, 28, -7)^{\mathsf{T}}. \tag{58}$$

Choosing $q_u = (16, 28, -7)^{\mathsf{T}}$ gives

$$q_a \cdot q_b = 1, \quad q_a \cdot q_u = -19, \quad q_b \cdot q_u = -5, \tag{59}$$

so the target triple is pairwise nonorthogonal. Exact arithmetic in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ verifies the full FOR: among all $\binom{53}{2} = 1378$ pairs, precisely the 102 pairs belonging to the 34 contexts are orthogonal, and no two rays are parallel. The accompanying certificate lists all rays, contexts, states, and normalized colorings, and includes a reproducible verifier.

D. State Counts and Separation from Operator Normalization

Exhaustive enumeration gives

$$\begin{aligned} &|\mathcal{S}(H_{53,34})| = 224, \qquad |\mathcal{S}_{\text{ext}}(H_{53,34})| = 128, \\ &N_c = 128 \quad \text{(768 labeled strong 3-colorings)}. \end{aligned} \tag{60}$$

The restrictions of all two-valued states to U are

$(s(a), s(b), s(u))$	multiplicity
$(1, 0, 0)$	80
$(0, 1, 0)$	80
$(0, 0, 1)$	64

(61)

There are exactly two independent chromatic admissible triples, $\{a, b, u\}$ and $\{a', b', u\}$; both are normalized on every two-valued state. Table IV records the remaining chromatic census counts for this configuration. The total of 25 non-context complete rainbow triples consists of these two independent triples and 23 partially nonorthogonal triples. No minimality claim is made for the 53 rays or 34 contexts.

The normalization in Eq. (53) has no affine certificate. The context incidence matrix has rank 34, and adjoining the row χ_U^{T} raises its rank to 35. The geometric separation is also explicit: for the Born preparation along q_c,

$$\begin{aligned} p_c(a) &= p_c(b) = 0, \\ p_c(u) &= \frac{|q_c \cdot q_u|^2}{\|q_c\|^2 \|q_u\|^2} = \frac{44^2}{2 \cdot 33^2} = \frac{8}{9}. \end{aligned} \tag{62}$$

Thus every classical two-valued state, and every convex mixture of them, gives total one on U, while this quantum state gives 8/9. Two-valued state admissibility therefore does not imply an affine or operator normalization identity.

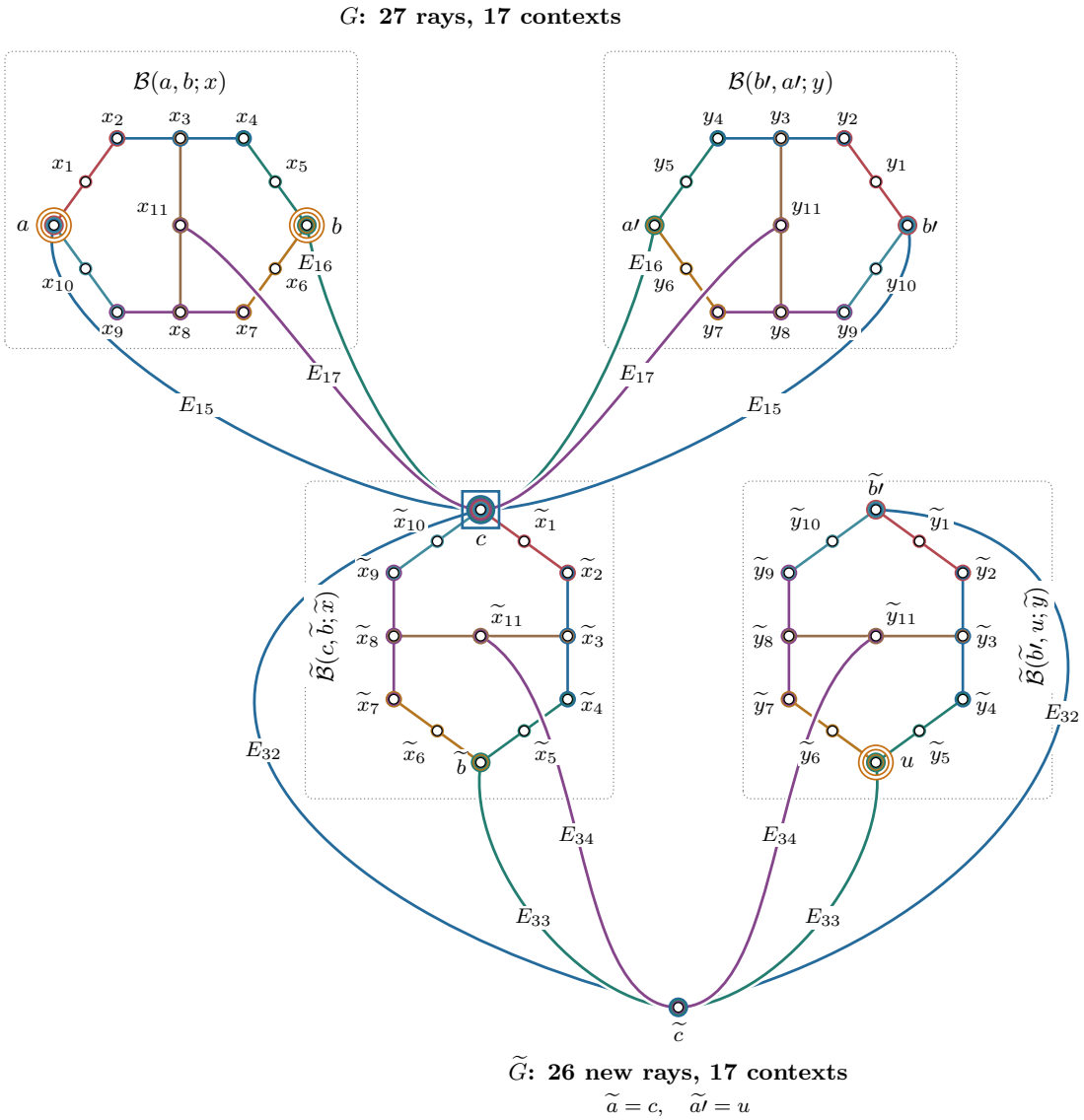


FIG. 3. The hypergraph $H_{53,34}$, with all 53 rays and 34 contexts. Dotted frames delimit seven-context Specker bugs; frame boundaries are not contexts. In G, the connecting contexts are $E_{15} = \{a, c, b'\}$, $E_{16} = \{b, c, a'\}$, and $E_{17} = \{c, x_{11}, y_{11}\}$; E_{32}, E_{33}, E_{34} are their counterparts in $\widetilde{G}$. Repeated labels identify both branches of each connecting context. The curves $E_{15}, E_{16}, E_{17}, E_{33}, E_{34}$ are smooth through their middle vertices. The upper bug layouts and E_{15}, E_{16}, E_{17} are mirror-symmetric about the vertical through c; E_{33} and E_{34} are individually symmetric about the vertical through $\widetilde{c}$, where they share a horizontal tangent. Crossings without a vertex symbol are not vertices. The square marks the unique shared ray $c = \widetilde{a}$, giving $27 + 27 - 1 = 53$ rays and $17 + 17 = 34$ contexts. Concentric rings record incident contexts; additional double outer outlines mark $U = \{a, b, u\}$, where $u = \widetilde{a'}$. On every two-valued state s, G enforces $s(a) + s(b) + s(c) = 1$, while $\widetilde{G}$ enforces $s(c) = s(u)$. Substitution yields $s(a) + s(b) + s(u) = 1$, proving two-valued state admissibility of U.

E. Parameterized Maximal Operators for Both Admissible Triples

Choose color labels so that $\kappa(a) = 0$, $\kappa(b) = 1$, and $\kappa(u) = 2$. This is possible because U is rainbow in every coloring. The FOR gives the explicit weighted operator

$$O_U = \frac{c_0}{3}\begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix} + \frac{c_1}{3}\begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & 1 \\ -1 & 1 & 1 \end{pmatrix} + \frac{c_2}{1089}\begin{pmatrix} 256 & 448 & -112 \\ 448 & 784 & -196 \\ -112 & -196 & 49 \end{pmatrix}. \tag{63}$$

With $t = c_0 + c_1 + c_2$, its characteristic polynomial is Eq. (23), where

$$\begin{aligned} b &= \tfrac{8}{9}c_0c_1 + \tfrac{2906}{3267}c_0c_2 + \tfrac{3242}{3267}c_1c_2, \\ \delta &= \tfrac{64}{81}c_0c_1c_2. \end{aligned} \tag{64}$$

Thus the exact three-parameter equality is

$$\max_{\rho}\big[c_0p_\rho(a) + c_1p_\rho(b) + c_2p_\rho(u)\big] = F(t, b, \delta). \tag{65}$$

Equations (24) and (18) give all eigenvalues and their spectral projections. For $U' = (a', b', u)$, independently relabel its three colors in the displayed order. The same equality holds after replacing b by

$$b' = \tfrac{8}{9}c_0c_1 + \tfrac{2929}{3267}c_0c_2 + \tfrac{3265}{3267}c_1c_2; \tag{66}$$

t and δ are unchanged. These coefficients follow from the rotated rays in Eq. (56) and the same q_u.

For equal unit color values, the first triple has eigenvalues

$$0.57095612, \qquad 0.91252305, \qquad 1.51652083,$$

the roots of $9801\lambda^3 - 29403\lambda^2 + 27156\lambda - 7744 = 0$. A normalized maximizing vector is approximately $(0.35550453, -0.92079461, 0.16047992)^{\mathsf{T}}$. The second triple has maximum 1.47574600. At these unit values, all classical two-valued states and their mixtures give expectation one on each target, while quantum preparations attain the stated larger values as well as values below one. The 8/9 preparation in Eq. (62) is therefore an explicit separation, but not an extremal one.

F. Sharp Classical Comparison for the Pasted Construction

For each of $U = (a, b, u)$ and $U' = (a', b', u)$, every two-valued state selects exactly one target vertex, and each of the three one-hot restrictions occurs. Hence Eq. (30) is the sharp classical comparison for the three-parameter quantum equalities in Section IV E, on both the full and extendable state families. In particular, at unit color values,

$$\begin{aligned} E_{\mathrm{cl}}(U) &= 1, \\ E_{\mathrm{q}}(U) &\in [0.57095612, 1.51652083]. \end{aligned} \tag{67}$$

The endpoints are rounded spectral values; the characteristic equality in Section IV E defines them exactly. Both endpoints separate the quantum and classical models. The classical equality follows structurally from the pasted nonseparating gadgets, so the finite enumeration is needed to certify the complete state inventory and the other target relations, rather than to replace that proof. The full comparison includes all 428 counted size-two and size-three paired relations and all four twin differences, as well as the single-set targets; its parameter dependence is stored as finite integer envelopes.

V. CHROMATIC ADMISSIBLE CONTEXTS AND COLOR-MULTISET EQUIVALENCE IN $\mathbb{R}^3$

A. Chromatic Admissibility and Operator Rigidity

Before exhibiting chromatic admissible contexts in $\mathbb{R}^3$, we distinguish their state-restricted normalization from an affine or operator identity.

Proposition 9 (Operator and Affine Rigidity of Complete d-Tuples)**.** *Let $H = (V, E)$ be a d-uniform hypergraph admitting a FOR $v \mapsto q_v \in \mathbb{C}^d$ with rank-one projectors $P_v = q_v q_v^\dagger/\|q_v\|^2$. If a d-element subset $U = \{u_1, \dots, u_d\} \subseteq V$ satisfies $\sum_{u\in U} P_u = I_d$ (in particular, if $\chi_U = M_H^{\mathsf{T}}\lambda$ with $\mathbf{1}_n^{\mathsf{T}}\lambda = 1$), then $q_{u_a} \perp q_{u_b}$ for all $1 \le a < b \le d$. Consequently, no set of d nonorthogonal rays can sum to I_d or be forced to sum to 1 by a linear combination of context equations.*

Proof. For each $a \in \{1, \dots, d\}$, conjugating $I_d - P_{u_a} = \sum_{b\neq a} P_{u_b}$ by the unit vector $\hat{q}_{u_a} = q_{u_a}/\|q_{u_a}\|$ gives

$$0 = \langle \hat{q}_{u_a}, (I_d - P_{u_a})\hat{q}_{u_a}\rangle = \sum_{b\neq a} \big|\langle \hat{q}_{u_a}, \hat{q}_{u_b}\rangle\big|^2,$$

so $\langle q_{u_a}, q_{u_b}\rangle = 0$ for all $b \neq a$. If $\chi_U = M_H^{\mathsf{T}}\lambda$ with $\mathbf{1}_n^{\mathsf{T}}\lambda = 1$, summing $\sum_{v\in C_i} P_v = I_d$ with weights λ_i yields $\sum_{u\in U} P_u = (\mathbf{1}_n^{\mathsf{T}}\lambda)I_d = I_d$, forcing mutual orthogonality. □

Proposition 9 excludes operator completeness and an affine normalization certificate for any d-tuple containing a nonorthogonal pair. It does not exclude normalization on a specified family of two-valued states. In particular, chromatic admissibility imposes Eq. (11) only on $\mathcal{S}_{\mathrm{ext}}(H)$. Whether this identity extends to all of $\mathcal{S}(H)$ is a separate question; the proposition alone does not imply that a violating nonextendable state must exist. The construction of Section IV supplies an explicit example normalized on all two-valued states but not on all Born states.

A sufficient additional hypothesis for lifting every identity on two-valued states to an affine identity is

$$\operatorname{aff}\mathcal{S}(H) = \mathcal{A}(H). \tag{68}$$

Under this hypothesis, normalization on all of $\mathcal{S}(H)$ extends to $\mathcal{A}(H)$ and hence has a context-equation certificate. Consequently, in a faithful orthogonal representation, an independent chromatic admissible context must then violate normalization on some nonextendable two-valued state. Without Eq. (68), no such general necessity is asserted. The examples below distinguish the two state families directly through their enumerated state sums.

B. Non-Twin Color-Multiset-Equivalent Pairs of Size Two in $\mathbb{R}^3$ ($I_{16,9}, H_{17,10}, T_{19,11}$)

A faithful real three-dimensional hypergraph can support a non-twin color-multiset-equivalent pair of size $k = 2$

TABLE II. Unnormalized integer vectors in $\mathbb{R}^3$ for the 25-ray completed Yu–Oh configuration H^+_{YO} (rays $1,\dots,25$, used by $B_{13,7}$, $H_{17,10}$, and $H_{21,13}$) and for the 19-ray gadget $T_{19,11}$ (second tuple where different, plus tip rays $26, 27, 28$) [8, 18]. For $H_{20,12}$ (omitting $\{6, 13, 22\}$), taking $r_1 = (0, 1, 2)$, $r_3 = (0, 2, -1)$, $r_7 = (2, 0, 1)$, $r_9 = (1, 0, -2)$, $r_{10} = (2, -2, 1)$, $r_{11} = (1, -1, -2)$, $r_{13} = (1, 4, -2)$ and cross-product completions yields a strictly faithful integer representation with $\langle r_6, r_{13}\rangle = -3 \neq 0$.

i	r_i (H^+_{YO} / $T_{19,11}$)	i	r_i (H^+_{YO} / $T_{19,11}$)	i	r_i (H^+_{YO} / $T_{19,11}$)	i	r_i (H^+_{YO} / $T_{19,11}$)
1	$(0,1,1)$ / $(0,1,2)$	8	$(0,1,0)$	15	$(1,2,-1)$ / $(1,4,-1)$	22	$(1,1,2)$
2	$(1,0,0)$	9	$(1,0,-1)$	16	$(1,-2,1)$	23	$(1,2,1)$ / $(1,4,1)$
3	$(0,1,-1)$ / $(0,2,-1)$	10	$(1,-1,1)$ / $(2,-1,2)$	17	$(1,1,-2)$	24	$(1,-1,2)$ / $(4,-2,5)$
4	$(1,1,0)$ / $(1,2,0)$	11	$(1,-1,-1)$ / $(2,-1,-2)$	18	$(1,-1,-2)$ / $(4,-2,-5)$	25	$(2,-1,-1)$
5	$(0,0,1)$	12	$(1,1,1)$	19	$(2,1,-1)$	26	$(2,-41,18)$ ($T_{19,11}$)
6	$(1,-1,0)$ / $(2,-1,0)$	13	$(1,1,-1)$	20	$(2,-1,1)$	27	$(200,-82,-209)$ ($T_{19,11}$)
7	$(1,0,1)$	14	$(2,1,1)$ / $(5,2,4)$	21	$(1,-2,-1)$	28	$(241,82,160)$ ($T_{19,11}$)

without an affine certificate. This does not contradict the prohibition of size-two real affine pseudocontexts in Ref. [2]: the color relation is required only on $\mathcal{S}_{\mathrm{ext}}(H)$. The construction combines a linear context identity with a non-context complete rainbow triple.

Theorem 10 (Non-Twin Color-Multiset Equivalence of Size Two in $\mathbb{R}^3$)**.** *In the faithful real three-dimensional hypergraphs $I_{16,9}$ (16 rays, 9 contexts, $N_c = 6$, $|\mathcal{S}_{\mathrm{ext}}|/|\mathcal{S}| = 12/22$) and $H_{17,10}$ (17 rays, 10 contexts, $N_c = 4$, $|\mathcal{S}_{\mathrm{ext}}|/|\mathcal{S}| = 10/18$), the nonorthogonal pairs*

$$\begin{aligned} A_1 &= \{2,4\}, \quad B_1 = \{14,19\}, \\ A_2 &= \{4,8\}, \quad B_2 = \{15,23\} \end{aligned} \tag{69}$$

have no chromatic twins between A_r and B_r, yet satisfy

$$\begin{aligned} \{\{c(2),c(4)\}\} &= \{\{c(14),c(19)\}\}, \\ \{\{c(4),c(8)\}\} &= \{\{c(15),c(23)\}\} \end{aligned} \qquad \forall\, c \in \mathcal{C}_3. \tag{70}$$

Similarly, in $T_{19,11}$ (19 rays, 11 contexts, $N_c = 7$, $|\mathcal{S}_{\mathrm{ext}}|/|\mathcal{S}| = 14/25$, Fig. 4(b)), the partially nonorthogonal tip triple $U_{\mathrm{tip}} = \{26, 27, 28\}$ (with $26 \perp 27$, $26 \perp 28$, $27 \not\perp 28$) is rainbow in every 3-coloring, and the nonorthogonal pairs $A_3 = \{3, 10\}$, $B_3 = \{24, 26\}$ satisfy $\{\{c(3), c(10)\}\} = \{\{c(24), c(26)\}\}$ for all $c \in \mathcal{C}_3(T_{19,11})$.

Proof. Both $I_{16,9}$ and $H_{17,10}$ contain the Specker-bug contexts $E_1 = \{1,2,3\}$, $E_2 = \{7,8,9\}$, $E_4 = \{2,8,5\}$, $E_8 = \{3,11,14\}$, $E_9 = \{7,11,15\}$, $E_{11} = \{9,10,23\}$, $E_{12} = \{1,10,19\}$ together with the two roof contexts $E_{10} = \{4,11,24\}$ and $E_{13} = \{4,10,18\}$ ($H_{17,10}$ also includes $E_3 = \{4,5,6\}$). On the Specker bug, every 3-coloring satisfies the universally rainbow condition $c(10) \neq c(11)$. Since 4 is adjacent to 11 in E_{10} and to 10 in E_{13}, $c(4) \neq c(11)$ and $c(4) \neq c(10)$. Thus $\{4, 10, 11\}$ is a partially nonorthogonal complete rainbow triple: $\{c(4), c(10), c(11)\} = \{0, 1, 2\}$ for all $c \in \mathcal{C}_3$, or equivalently by Theorem 3(1),

$$s(4) + s(10) + s(11) = 1 \qquad \forall\, s \in \mathcal{S}_{\mathrm{ext}}. \tag{71}$$

Meanwhile, on *every* two-valued state $s \in \mathcal{S}$, the context combination $E_8 + E_{12} - E_1$ gives

$$\begin{aligned} \big(s(3) + s(11) + s(14)\big) + \big(s(1) + s(10) + s(19)\big) \\ - \big(s(1) + s(2) + s(3)\big) = 1, \end{aligned} \tag{72}$$

so $s(14) + s(19) - s(2) = 1 - s(10) - s(11)$. Substituting Eq. (71) into Eq. (72) yields $s(2) + s(4) = s(14) + s(19)$ for all $s \in \mathcal{S}_{\mathrm{ext}}$, and symmetric combination $E_9 + E_{11} - E_2$ yields $s(4) + s(8) = s(15) + s(23)$ for all $s \in \mathcal{S}_{\mathrm{ext}}$. By Theorem 3(3), Eq. (70) holds in every 3-coloring. In $H_{17,10}$ [Eq. (14) of Ref. [9]], $(c(2), c(4))$ and $(c(14), c(19))$ equal $(0, 0)$ and $(0, 0)$ in c_1, c_3, $(0, 1)$ and $(1, 0)$ in c_2, and $(0, 1)$ and $(0, 1)$ in c_4, confirming the absence of chromatic twins between A_1 and B_1.

In $T_{19,11}$ [Fig. 4(b)], the two upper contexts $\{14, 27, 26\}$ and $\{18, 28, 26\}$ together with the universally rainbow pair $c(14) \neq c(18)$ force $18 \sim_c 27$, $14 \sim_c 28$, and $\{c(26), c(27), c(28)\} = \{0, 1, 2\}$. Combining $\{3, 11, 14\} + \{4, 10, 18\} - \{4, 11, 24\}$ with $c(18) = c(27)$ and $\{14, 27, 26\}$ gives $s(3) + s(10) = s(24) + s(26)$ on $\mathcal{S}_{\mathrm{ext}}(T_{19,11})$. □

C. Chromatic Admissible Contexts in $H_{20,12}$ and $H_{21,13}$

Finally, we turn to Class I of Definition 2: a single triple $U = \{u_1, u_2, u_3\}$ of *pairwise nonorthogonal* rays in $\mathbb{R}^3$ ($q_{u_a} \not\perp q_{u_b}$ for all $a \neq b$) that is chromatically complete ($\{c(u_1), c(u_2), c(u_3)\} = \{0, 1, 2\}$) in *every* strong 3-coloring $c \in \mathcal{C}_3(H)$.

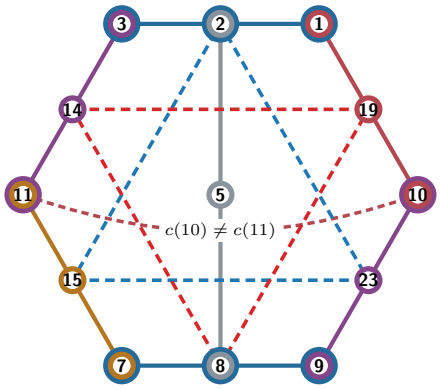
Lemma 11 (Structural Color Propagation in $H_{21,13}$)**.** *Every strong 3-coloring gives distinct colors to the hubs $10, 11, 13$ and to each of the triples $\{1, 4, 7\}$ and $\{3, 6, 9\}$. There are exactly two normalized strong 3-colorings, representing all 12 labeled colorings.*

Proof. Suppose first that hubs $10, 11$ share color t. Vertices $1, 3$ both avoid t, so their context forces $c(2) = t$. Similarly, vertices $7, 9$ both avoid t, forcing $c(8) = t$, contrary to the context $\{2, 5, 8\}$. Equating hubs $10, 13$ similarly forces $c(5) = c(8)$, while equating $11, 13$ forces $c(2) = c(5)$. Thus the hub triple is rainbow.

Adjacency to two hubs now forces

$$c(1) = c(11), \quad c(4) = c(13), \quad c(7) = c(10),$$

(a) Specker bug $B_{13,7}$: $c(10) \neq c(11)$ and $A = \{2, 15, 23\}$, $B = \{8, 14, 19\}$



(b) $T_{19,11}$: rainbow $\{26, 27, 28\}$ and size-two $A_3 = \{3, 10\}$, $B_3 = \{24, 26\}$

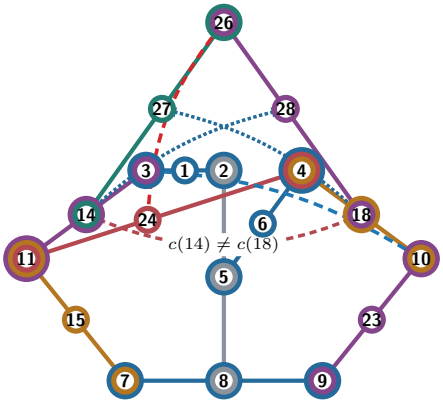


FIG. 4. Three-dimensional gadgets in $\mathbb{R}^3$ (coordinates in Table II): (a) Specker bug $B_{13,7}$ with universally rainbow pair $c(10) \neq c(11)$ and size-three affine pseudocontexts $A = \{2, 15, 23\}$ (blue dashed triangle) and $B = \{8, 14, 19\}$ (red dashed triangle); (b) $T_{19,11}$ with universally rainbow pair $c(14) \neq c(18)$, chromatic twins $14 \sim_c 28$ and $18 \sim_c 27$, complete rainbow tip triple $\{26, 27, 28\}$, and the non-twin color-multiset-equivalent pair $A_3 = \{3, 10\}$ (blue dashed arc) and $B_3 = \{24, 26\}$ (red dashed arc).

so $\{1, 4, 7\}$ is rainbow. Arrange the three base contexts as rows of

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

Every row contains each color once. The first column is rainbow by the preceding argument, and the middle column is an actual context. Counting occurrences of each color therefore leaves exactly one occurrence in the last column, proving that $\{3, 6, 9\}$ is rainbow.

Normalize the middle column to $(0, 1, 2)$. The first column is a permutation avoiding the middle-column color in each row, so it is one of the two derangements of three colors. The last column, the hubs, and every remaining vertex are then uniquely determined. Both derangements extend to all contexts. Hence there are two normalized colorings, and their six color permutations exhaust all 12 labeled strong colorings. □

Theorem 12 (Sharp Minimality and Transversal Classification in Yu–Oh Restrictions)**.** *Consider the* 13*-context,* 21*-ray Yu–Oh restriction* $H_{21,13}$ *in* $\mathbb{R}^3$ *(Table II and Fig. 5) obtained by omitting the single diagonal sector* $h_0 = 12$ *from* H_{YO}^{+} *[8, 9].*

1. Sharp threshold at 12 contexts ($H_{20,12}$): *Among all* $2^{13} = 8{,}192$ *context subhypergraphs of* $H_{21,13}$*, none with* $|E| \leq 11$ *contexts possesses a pairwise-nonorthogonal complete rainbow triple. At* $|E| = 12$*, deleting a single roof context from one diagonal sector—for instance omitting* $\{6, 13, 22\}$ *to form* $H_{20,12}$ *(*$|V| = 20$*,* $|E| = 12$*,* $N_c = 4$*,* $|\mathcal{S}_{\mathrm{ext}}|/|\mathcal{S}| = 10/25$*)—yields the minimal configuration, which already forces two pairwise-nonorthogonal complete rainbow triples:*

$$U_1^{(20)} = \{15, 18, 20\}, \qquad U_2^{(20)} = \{19, 21, 24\}. \tag{73}$$

2. Even-transversal classification in $H_{21,13}$ ($|V| = 21$, $|E| = 13$, $N_c = 2$, $|\mathcal{S}_{\mathrm{ext}}|/|\mathcal{S}| = 6/21$): *Each vertex* $v \in V(H_{21,13})$ *has a bivariate color signature* $\sigma(v) = (c_1(v), c_2(v)) \in \{0, 1, 2\}^2$ *across the two normalized* 13×3 *Context–Color Matrices [Eq. (16) of Ref. [9]], partitioning* $V(H_{21,13})$ *into the* 3×3 *signature grid:*

$c_2 \backslash c_1$	$\mathbf{c}_0 = 0$	$\mathbf{c}_1 = 1$	$\mathbf{c}_2 = 2$
0	$\{2, 14\}$	$\{6\}$	$\{7, 10, 20, 24\}$
1	$\{4, 13, 15, 19\}$	$\{8, 23\}$	$\{3\}$
2	$\{9\}$	$\{1, 11, 18, 21\}$	$\{5, 22\}$

(74)

A triple $U \subseteq V(H_{21,13})$ *is rainbow in every* 3*-coloring if and only if* U *is a* permutation transversal *[19] of Eq. (74), selecting one vertex from each row and one from each column (*$\sum_{\pi \in S_3} \prod_{i=0}^{2} |F_{i,\pi(i)}| = 97$ *total transversals, i.e.,* 13 *contexts plus* 84 *non-context rainbow triples). The three odd permutations* $\pi \in S_3$ *contain* 0 *pairwise-nonorthogonal triples, whereas the three even (cyclic) permutations yield* exactly 21 chromatic admissible contexts*:*

- Singleton cyclic transversal $F_{0,2} \times F_{1,0} \times F_{2,1}$ (1 triple):

$$U_{\mathrm{cop}} = \{3, 6, 9\}, \tag{75}$$

whose rays $r_3 = (0, 1, -1)$*,* $r_6 = (1, -1, 0)$*,* $r_9 = (1, 0, -1)$ *are not only pairwise nonorthogonal (*$\langle r_3, r_6 \rangle = -1$*,* $\langle r_3, r_9 \rangle = 1$*,* $\langle r_6, r_9 \rangle = 1$*) but* coplanar *(*$r_3 + r_6 - r_9 = \mathbf{0}$*, spanning the*

plane orthogonal to omitted hub $r_{12} = (1, 1, 1)$ *so* $\operatorname{rank}(P_3 + P_6 + P_9) = 2 < 3$*).*

- Main diagonal transversal $F_{0,0} \times F_{1,1} \times F_{2,2}$ (4 of 8 triples):

$$\{2, 22, 23\}, \quad \{5, 14, 23\}, \quad \{8, 14, 22\}, \quad \{14, 22, 23\}. \tag{76}$$

- Cyclic transversal $F_{0,1} \times F_{1,2} \times F_{2,0}$ (16 of 64 triples):

$$\begin{aligned} &\{1, 4, 7\},\ \{1, 15, 24\},\ \{4, 20, 21\},\ \{7, 18, 19\},\\ &\{10, 11, 13\},\ \{10, 15, 21\},\ \{11, 19, 20\},\ \{13, 18, 24\},\\ &\{15, 18, 20\},\ \{15, 18, 24\},\ \{15, 20, 21\},\ \{15, 21, 24\},\\ &\{18, 19, 20\},\ \{18, 19, 24\},\ \{19, 20, 21\},\ \{19, 21, 24\}. \end{aligned} \tag{77}$$

In particular, $U_{\text{core}} = \{1, 4, 7\}$ *consists of three core rays* $r_1 = (0, 1, 1)$, $r_4 = (1, 1, 0)$, $r_7 = (1, 0, 1)$ *at pairwise* $60°$*, and* $U_{\text{hub}} = \{10, 11, 13\}$ *consists of the three retained diagonal hubs* $r_{10} = (1, -1, 1)$, $r_{11} = (1, -1, -1)$, $r_{13} = (1, 1, -1)$.

3. Chromatic twin classes of size four ($k = 4 > d = 3$): *All four rays within each of the three 4-element signature fibers* $F_{0,2} = \{7, 10, 20, 24\}$, $F_{1,0} = \{4, 13, 15, 19\}$, *and* $F_{2,1} = \{1, 11, 18, 21\}$ *of Eq. (74) are pairwise nonorthogonal in* $\mathbb{R}^3$ *and share the exact same color in every 3-coloring of* $H_{21,13}$.

Proof. By Lemma 11, any strong 3-coloring $c \in \mathcal{C}_3(H_{21,13})$ is obtained by applying a color permutation $\tau \in S_3$ to either c_1 or c_2. Hence $\{c(u_1), c(u_2), c(u_3)\} = \{0, 1, 2\}$ holds for all $c \in \mathcal{C}_3(H_{21,13})$ iff $\{c_1(u_1), c_1(u_2), c_1(u_3)\} = \{0, 1, 2\}$ and $\{c_2(u_1), c_2(u_2), c_2(u_3)\} = \{0, 1, 2\}$, meaning that the three signatures $\sigma(u_1), \sigma(u_2), \sigma(u_3)$ have distinct first coordinates and distinct second coordinates—the definition of a permutation transversal of Eq. (74). Exact inner-product checks on the 97 transversals give the listed geometric classification and twin classes. Part (1) is a computer-assisted minimality result: all 2^{13} context subhypergraphs were enumerated, their strong colorings computed, and their independent complete triples tested. The elementary universal-completeness and coloring-count arguments are supplied separately by Lemma 11. □

Over all 21 ordinary two-valued states $s \in \mathcal{S}(H_{21,13})$, the sum $s(u_1) + s(u_2) + s(u_3)$ on $U_{\text{core}} = \{1, 4, 7\}$ takes values $0, 1, 2$ with multiplicities $6, 12, 3$, and on $U_{\text{cop}} = \{3, 6, 9\}$ takes values $0, 1, 2$ with multiplicities $3, 12, 6$; on the 6 extendable states $\mathcal{S}_{\text{ext}}(H_{21,13})$ it is identically 1.

D. Color-Weighted Quantum Maxima for the Chromatic Targets

For each ordered rainbow triple in Table III, choose color labels so that its three weights are (c_0, c_1, c_2). The table gives an exact parameterized eigensystem through

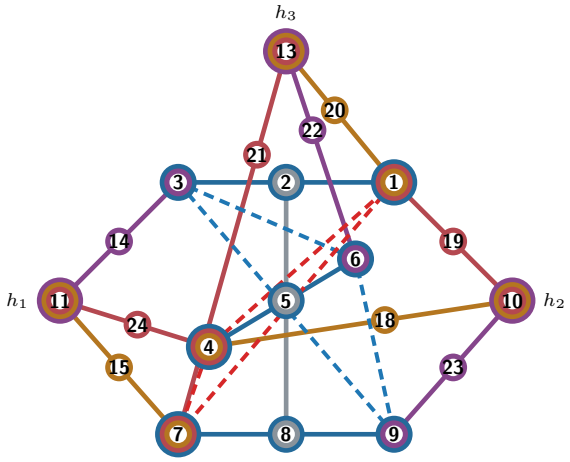


FIG. 5. The 21-ray, 13-context Yu–Oh restriction $H_{21,13}$ in $\mathbb{R}^3$ (all 13 contexts drawn as straight lines; coordinates in Table II). Highlighted are two of the 21 chromatic admissible contexts (Theorem 12): the coplanar triple $U_{\text{cop}} = \{3, 6, 9\}$ (blue dashed triangle, $r_3 + r_6 - r_9 = \mathbf{0}$) and the $60°$ basis triple $U_{\text{core}} = \{1, 4, 7\}$ (red dashed triangle), both satisfying $\{c(u_1), c(u_2), c(u_3)\} = \{0, 1, 2\}$ in every 3-coloring of $H_{21,13}$.

TABLE III. Exact characteristic-polynomial coefficients for representative ordered chromatic admissible triples. The same Gram-minor construction gives the coefficients of every other triple in Eqs. (76)–(77).

Ordered triple	h_{01}	h_{02}	h_{12}	g
$H_{21,13}$: $(1, 4, 7)$	3/4	3/4	3/4	1/2
$H_{21,13}$: $(3, 6, 9)$	3/4	3/4	3/4	0
$H_{21,13}$: $(10, 11, 13)$	8/9	8/9	8/9	16/27
$H_{20,12}$: $(15, 18, 20)$	131/135	1573/1575	913/945	529/567
$H_{20,12}$: $(19, 21, 24)$	4469/4725	134/135	314/315	529/567

Eqs. (22)–(24): its maximal expectation is $F(c_0 + c_1 + c_2, b, g c_0 c_1 c_2)$, where $b = h_{01} c_0 c_1 + h_{02} c_0 c_2 + h_{12} c_1 c_2$. The $H_{20,12}$ rows use the separate faithful coordinates specified in Table II, rather than the coordinates of $H_{21,13}$.

For the core and coplanar triples, the operators themselves are particularly transparent:

$$\begin{aligned} O_{\text{core}} &= \frac{1}{2} \begin{pmatrix} c_1 + c_2 & c_1 & c_2 \\ c_1 & c_0 + c_1 & c_0 \\ c_2 & c_0 & c_0 + c_2 \end{pmatrix}, \\ O_{\text{cop}} &= \frac{1}{2} \begin{pmatrix} c_1 + c_2 & -c_1 & -c_2 \\ -c_1 & c_0 + c_1 & -c_0 \\ -c_2 & -c_0 & c_0 + c_2 \end{pmatrix}. \end{aligned} \tag{78}$$

Writing $t = c_0 + c_1 + c_2$ and $b = \frac{3}{4}(c_0 c_1 + c_0 c_2 + c_1 c_2)$, the coplanar eigensystem has the fixed zero eigenvector

$(1,1,1)^{\mathsf{T}}/\sqrt{3}$ and two remaining eigenvalues

$$\lambda_{\pm} = \tfrac{1}{2}\left(t \pm \sqrt{t^2 - 4b}\right),$$
$$\max_{\rho}\left[c_0 p_\rho(3) + c_1 p_\rho(6) + c_2 p_\rho(9)\right] = \max\{0, \lambda_+\}. \tag{79}$$

The other eigenspaces are obtained from Eq. (18). This formula includes negative color values, for which the fixed zero eigenvalue can be the maximum.

At $\mathbf{c} = \mathbf{1}$, put $n = (1,1,1)^{\mathsf{T}}/\sqrt{3}$. Then

$$\begin{aligned} O_{\text{core}} &= 2nn^{\mathsf{T}} + \tfrac{1}{2}(I_3 - nn^{\mathsf{T}}), \\ O_{\text{cop}} &= \tfrac{3}{2}(I_3 - nn^{\mathsf{T}}), \\ O_{\text{hub}} &= \tfrac{1}{3}nn^{\mathsf{T}} + \tfrac{4}{3}(I_3 - nn^{\mathsf{T}}). \end{aligned} \tag{80}$$

Their maxima are therefore 2, 3/2, and 4/3, respectively. The maximizing preparations are n for the core and any unit vector in $n^{\perp}$ for the other two. These are quantum maxima for the fixed FOR; the normalization value one on extendable states remains a separate statement.

The smaller chromatic relations have explicit two-ray eigensystems as well. For any ordered pair of distinct rays (u, v), put $a = c_{\kappa(u)}$, $b = c_{\kappa(v)}$, and $r = |\langle \hat{q}_u, \hat{q}_v \rangle|^2$. In $\mathbb{C}^3$,

$$\operatorname{spec}(aP_u + bP_v) = \left\{0, \frac{a + b \pm \sqrt{(a-b)^2 + 4abr}}{2}\right\}. \tag{81}$$

Thus its maximum is the larger of zero and the plus branch. For a chromatic twin pair, $a = b$ and the nonzero branches are $a(1 \pm \sqrt{r})$. For the non-twin paired sets in Section V B, use Eq. (38); their difference is traceless by color-multiset equivalence, and Eq. (39) gives its exact maximum. This applies to a fixed coloring and its actual color weights, without promoting a chromatic relation to an affine certificate.

E. Classical Comparisons for the Logical and Yu–Oh Targets

The two independent triples of $H_{20,12}$ have the seven nonzero binary restrictions on their 25 two-valued states. For either ordered triple, with colors normalized to $(0,1,2)$, the sharp full-state bounds are the minimum and maximum of

$$\{c_0, c_1, c_2, c_0 + c_1, c_0 + c_2, c_1 + c_2, c_0 + c_1 + c_2\}. \tag{82}$$

On the ten extendable states the bounds instead come from $\{c_0, c_1, c_2\}$. At unit values, both triples have full classical interval $[1,3]$ and quantum interval approximately $[0.72960967, 1.23483806]$ in the stated faithful representation. The quantum minimum therefore violates the full classical lower bound; the quantum maximum exceeds one only relative to the extendable-state normalization.

For $H_{21,13}$, each of the core and coplanar triples has all binary restrictions except $(1,1,1)$. Their exact full classical envelope is Eq. (46). Thus their unit quantum intervals, $[1/2, 2]$ and $[0, 3/2]$, are contained in the full classical interval $[0,2]$. In fact no choice of real color values on either of these triples separates all two-valued states: their classical projected polytope is

$$\{x \in [0,1]^3 : x_0 + x_1 + x_2 \le 2\},$$

and every Born probability vector lies in it, by the displayed spectral upper bounds and $0 \le p_\rho(v) \le 1$.

The hub triple behaves differently. Its full-state restrictions are $000, 100, 010, 001$, giving

$$\begin{aligned} \beta_{\mathcal{S}}^{-}(\mathbf{c}) &= \min\{0, c_0, c_1, c_2\}, \\ \beta_{\mathcal{S}}^{+}(\mathbf{c}) &= \max\{0, c_0, c_1, c_2\}. \end{aligned} \tag{83}$$

At unit values its full classical interval is $[0,1]$, whereas its quantum maximum is 4/3. This is a separation from all two-valued states, not only from the six extendable states.

The same calculation distinguishes the smaller logical gadgets. The pair $\{14, 18\}$ in $D_{16,9}$ has all four binary restrictions on $\mathcal{S}$, but only $00, 10, 01$ on $\mathcal{S}_{\text{ext}}$. Its quantum unit maximum 49/45 therefore exceeds the extendable bound one while remaining within the full bound two. In $I_{16,9}$ and $H_{17,10}$, the original bug tips $\{10, 11\}$ are exclusive on all two-valued states, and their quantum maxima are approximately 1.27216553 and 4/3, respectively. The first value uses the integer FOR for panel (d) in the companion's vector inventory [18], explicitly included in the computational inventory; the graph's state space does not change when the FOR is changed. In $T_{19,11}$, the pair $\{9, 26\}$ is exclusive on all 25 two-valued states and has quantum maximum 1.25241492, even though the particular relation $\{14, 18\}$ is purely chromatic. The complete comparison retains every counted rainbow pair, twin difference, and paired relation, so these representative comparisons do not replace the full inventory.

F. Two-Valued Admissibility Contexts in $\mathbb{R}^3$

Can an independent triple be normalized on *every* two-valued state of a faithfully represented, 3-colorable hypergraph in $\mathbb{R}^3$? The answer is affirmative. The mechanism below has two separate ingredients: a binary implication across two contexts, and a geometric realization of the gadgets supplying its exclusivity assumptions. Figure 6 displays the implication in two stages.

Lemma 13 (Roof Twins)**.** *Let H contain the contexts $\{a, x, y'\}$ and $\{a, y, x'\}$. Suppose*

$$s(x) + s(y) \le 1, \qquad s(x') + s(y') \le 1 \tag{84}$$

for every $s \in \mathcal{S}(H)$. Then every such state satisfies

$$s(x) = s(x'), \qquad s(y) = s(y'), \tag{85}$$

and consequently

$$\begin{aligned} s(a) + s(x) + s(y) &= 1, \\ s(a) + s(x') + s(y') &= 1. \end{aligned} \tag{86}$$

Proof. Subtracting the two context equations gives

$$\delta = s(x) - s(x') = s(y) - s(y') \in \{-1, 0, 1\}.$$

If $\delta = 1$, then $s(x) = s(y) = 1$; if $\delta = -1$, then $s(x') = s(y') = 1$. Both contradict Eq. (84). Hence $\delta = 0$, proving the twin identities. Substitution into either context equation gives Eq. (86). This step uses the binary range of s. □

Write $B(x, y; \ell)$ for a Specker bug with tips x, y and eleven new vertices $\ell + k$, where $k \in \{1, 2, 3, 5, 7, 8, 9, 14, 15, 19, 23\}$. Its seven contexts are

$$\begin{gathered} \{\ell+1, \ell+2, \ell+3\}, \quad \{\ell+7, \ell+8, \ell+9\}, \\ \{\ell+2, \ell+8, \ell+5\}, \\ \{\ell+3, y, \ell+14\}, \quad \{\ell+7, y, \ell+15\}, \\ \{\ell+9, x, \ell+23\}, \quad \{\ell+1, x, \ell+19\}. \end{gathered} \tag{87}$$

These contexts force $s(x) + s(y) \le 1$: if both tips were one, vertices $\ell+1, \ell+3, \ell+7, \ell+9$ would be zero, so the first two contexts would force $s(\ell+2) = s(\ell+8) = 1$, contradicting the third context. This is the only bug implication needed for the cascade.

Lemma 14 (Specker-Bug Coordinates)**.** *Let $u, v \in \mathbb{R}^3$ have equal nonzero norm and $0 < \langle \hat{u}, \hat{v} \rangle \le 1/3$, after reversing one vector if necessary. Set $e_1 = u + v$, $e_2 = u - v$, $e_3 = u \times v$, and*

$$\cos\phi = \sqrt{2 - \frac{\|e_1\|^2}{\|e_2\|^2}}, \qquad \sin\phi = \sqrt{\frac{\|e_1\|^2}{\|e_2\|^2} - 1}. \tag{88}$$

The orthonormal triad

$$\begin{aligned} f_{1,2} &= \tfrac{1}{\sqrt{2}}\big(\hat{e}_1 \pm (\cos\phi\, \hat{e}_2 + \sin\phi\, \hat{e}_3)\big), \\ f_3 &= f_1 \times f_2 \end{aligned} \tag{89}$$

gives the intended orthogonalities of Eq. (87). Explicitly, with $r_x = u$, $r_y = v$ and $q_k = r_{\ell+k}$, take

$$\begin{aligned} q_2 &= f_1, & q_8 &= f_2, & q_5 &= f_3, \\ q_1 &= f_1 \times u, & q_3 &= f_1 \times v, & & \\ q_9 &= f_2 \times u, & q_7 &= f_2 \times v, & & \\ q_{14} &= v \times q_3, & q_{15} &= v \times q_7, & & \\ q_{19} &= u \times q_1, & q_{23} &= u \times q_9. & & \end{aligned} \tag{90}$$

Faithfulness of a compound hypergraph additionally requires checking that all its rays are distinct and that no orthogonal pair is absent from its two-section graph.

Proof. The overlap bound makes both square roots real. The vectors e_1, e_2, e_3 are mutually orthogonal, so f_1, f_2, f_3 are orthonormal. Lagrange's identity gives

$$(f_i \times u) \cdot (f_i \times v) = u \cdot v - (f_i \cdot u)(f_i \cdot v) = 0,$$

because

$$(f_i \cdot u)(f_i \cdot v) = \tfrac{1}{8}(\|e_1\|^2 - \cos^2\phi\, \|e_2\|^2) = u \cdot v.$$

Thus $q_1 \perp q_3$ and $q_7 \perp q_9$; all remaining required orthogonalities follow directly from the cross products in Eq. (90). □

Theorem 15 (Roof-Twin Cascades in $\mathbb{R}^3$)**.** *Starting from the nine-context hypergraph $I_{16,9}$, define*

$$\begin{aligned} G_{27,16} &= I_{16,9} \cup B(18, 24; 100), \\ G_{41,25} &= G_{27,16} \cup \{\{14, 27, 26\}, \{18, 28, 26\}\} \\ &\qquad \cup B(27, 28; 200), \\ G_{41,26} &= G_{41,25} \cup \{\{4, 5, 105\}\}. \end{aligned}$$

All three admit FORs in $\mathbb{R}^3$ and strong 3-colorings. Their respective values of $(N_c, |\mathcal{S}_{\text{ext}}|, |\mathcal{S}|)$ are $(36, 54, 82)$, $(216, 216, 396)$, and $(48, 63, 107)$. Every two-valued state of $G_{27,16}$ satisfies $s(11) = s(18)$ and $s(10) = s(24)$. In each of the two 41-ray hypergraphs, every two-valued state satisfies

$$\begin{gathered} s(11) = s(18) = s(27), \qquad s(14) = s(28), \\ s(3) = s(26). \end{gathered} \tag{91}$$

In particular,

$$\begin{gathered} U_{\text{TVS}} = \{3, 27, 28\}, \\ s(3) + s(27) + s(28) = 1 \qquad (\forall s \in \mathcal{S}) \end{gathered} \tag{92}$$

is a pairwise-nonorthogonal two-valued state admissibility context in both $G_{41,25}$ and $G_{41,26}$.

Proof. Binary implication. The original bug $B(10, 11; 0)$ and the attached $B(18, 24; 100)$ supply the two exclusivity assumptions for the roof contexts $\{4, 11, 24\}$ and $\{4, 10, 18\}$. Lemma 13 gives $s(11) = s(18)$ and $s(10) = s(24)$. Context $E_8 = \{3, 11, 14\}$ therefore yields

$$s(3) + s(14) + s(18) = 1, \tag{93}$$

so $s(14) + s(18) \le 1$. The third bug supplies $s(27) + s(28) \le 1$. Applying the same lemma to $\{26, 14, 27\}$ and $\{26, 18, 28\}$ gives $s(14) = s(28)$ and $s(18) = s(27)$. Substitution into Eq. (93) proves Eq. (92); comparison with either second-roof context also gives $s(3) = s(26)$. Adding a context only restricts the state space, so these implications hold for both 41-ray cases. This proves universal normalization without enumerating states.

Faithful coordinates. Geometric realizations must be distinguished from the preceding binary argument. A parametrization of $I_{16,9}$ is

$$\begin{aligned} r_1 &= (0, 1, \alpha), & r_2 &= (1, 0, 0), \\ r_3 &= (0, \alpha, -1), & r_5 &= (0, 0, 1), \\ r_7 &= (1, 0, \beta), & r_8 &= (0, 1, 0), \\ r_9 &= (\beta, 0, -1), & r_{10} &= (1, -\alpha\beta, \beta), \\ r_{11} &= (\alpha\beta, -1, -\alpha), & r_4 &= r_{10} \times r_{11}, \\ r_{14} &= r_{11} \times r_3, & r_{15} &= r_7 \times r_{11}, \\ r_{19} &= r_1 \times r_{10}, & r_{23} &= r_{10} \times r_9, \\ r_{18} &= r_4 \times r_{10}, & r_{24} &= r_{11} \times r_4. \end{aligned} \tag{94}$$

At $(\alpha, \beta) = (9/32, 24/11)$,

$$\begin{aligned} \|r_4\|^2 &= (1 + \alpha^2)(1 + \beta^2)(1 + \alpha^2\beta^2), \\ \|r_4\| &= 45305/15488. \end{aligned} \tag{95}$$

Put $n = r_4/\|r_4\|$ and $R = nn^{\mathsf{T}} - [n]_\times$, where $[n]_\times w = n \times w$. This is a quarter-turn about n. Both tips lie in $n^\perp$, where $Rw = -n \times w$, so it sends the rays of r_{10}, r_{11} to those of r_{18}, r_{24}, respectively. Assigning $r_{100+k} = Rr_k$ for the eleven internal bug labels in Eq. (87) gives a rational FOR of $G_{27,16}$ (and integer representatives after rescaling).

At these same rational parameters, take $r_{26} = r_{14} \times r_{18}$, $r_{27} = r_{26} \times r_{14}$, $r_{28} = r_{26} \times r_{18}$ and apply Lemma 14 to the last two rays. Their squared normalized overlap is

$$\frac{21159411204096}{3980389958002705} < \frac{1}{9}.$$

This gives a FOR of $G_{41,25}$. The rotation and cross-product formulas establish all intended orthogonalities algebraically; outward-rounded interval evaluation at 70 decimal digits certifies every other inner product to be nonzero and every pair of rays to be distinct. The smallest certified absolute normalized nonedge inner product exceeds 0.00155.

For $G_{41,26}$, a simpler, separate realization uses $\alpha = \beta = 1$. Choose the ray representatives $r_4 = (1,1,0)$, $r_{14} = (2,1,1)$, $r_{18} = (1,-1,-2)$, $r_{24} = (1,-1,2)$. For the second bug use Eq. (90) with $f_1 = (1,1,\sqrt{2})$, $f_2 = (1,1,-\sqrt{2})$ (normalization is immaterial for rays). For the second roof use

$$\begin{aligned} r_{26} &= (-1,5,-3), \\ r_{27} &= (8,-5,-11), \\ r_{28} &= (-13,-5,-4). \end{aligned} \tag{96}$$

Their normalized overlap has magnitude $1/6$. For the third bug, Eq. (90) applies with

$$\begin{aligned} f_{1,2} = \sqrt{35}\,(3,0,-1) \\ \pm \left(\sqrt{15}\,(1,2,3) + 2(-1,5,-3)\right). \end{aligned} \tag{97}$$

Exact arithmetic in $\mathbb{Q}(\sqrt{2}, \sqrt{15}, \sqrt{35})$ verifies all 820 ray pairs: rays are distinct, and orthogonality is precisely adjacency in $G_{41,26}$. In particular,

$$r_3 \cdot r_{27} = 6, \quad r_3 \cdot r_{28} = -1, \quad r_{27} \cdot r_{28} = -35,$$

so the target triple is pairwise nonorthogonal. At these symmetric parameters, deleting $\{4,5,105\}$ leaves three unrecorded orthogonal pairs, namely $(4,5), (4,105), (5,105)$. Thus these symmetric coordinates are faithful for $G_{41,26}$, whereas $G_{27,16}$ and $G_{41,25}$ use the separate rational-parameter construction above.

Finite counts. Exhaustive enumeration, independently checked by a second implementation, gives the stated state and normalized-coloring counts. The coordinate and state certificate supplies the full contexts, ray coordinates, assignments, and verification routines. Enumeration certifies the counts and geometry; the two applications of Lemma 13 establish normalization on *all* two-valued states. □

The separation from affine and operator normalization can be seen directly. In the symmetric FOR of $G_{41,26}$, prepare $\psi = \hat{r}_3 = (0,1,-1)/\sqrt{2}$. Since $\|r_{27}\|^2 = \|r_{28}\|^2 = 210$, its Born probabilities give

$$\begin{aligned} p_\psi(3) + p_\psi(27) + p_\psi(28) &= 1 + \frac{36}{420} + \frac{1}{420} \\ &= \frac{457}{420} \neq 1. \end{aligned} \tag{98}$$

This weight lies in $\mathcal{W}(G_{41,26}) \subset \mathcal{A}(G_{41,26})$, while every convex combination of two-valued states has target sum one. Hence $\operatorname{aff} \mathcal{S}(G_{41,26}) \subsetneq \mathcal{A}(G_{41,26})$ and $P_3 + P_{27} + P_{28} \neq I_3$, in agreement with Proposition 9. The uniform weight $p(v) = 1/3$ satisfies the target normalization and every twin identity; it is not a separating witness. Indeed, any strong 3-coloring expresses it as the average of its three color-class states.

Proposition 16 (Binary Restrictions in the Yu–Oh Examples)**.** *In the fixed Yu–Oh representation, the unique triple of pairwise nonorthogonal rays satisfying* $\sum_{v \in U} s(v) \leq 1$ *on every state of* $H_{21,13}$ *is* $U = \{10, 11, 13\}$*. No proper context subhypergraph has such a triple. Moreover, impose* $s(a) + s(b) \leq 1$ *for the following pairs* (a, b)*:*

$$\begin{aligned} H_{20,12}: &\quad (18,24), (20,21), \\ H_{21,13}: &\quad (18,24), (15,21), (19,20). \end{aligned} \tag{99}$$

These filters retain exactly the original extendable states: 10 *of* 25 *states for* $H_{20,12}$*, and* 6 *of* 21 *for* $H_{21,13}$*. Thus the two, respectively* 21*, independent chromatic admissible triples are normalized on these restricted state families.*

Proof. Enumeration of the 21 states and the independent triples of $H_{21,13}$ gives the unique exclusive triple $\{10, 11, 13\}$. For each of the 13 single-context deletions, the certificate supplies a state with hub sum greater than one. Every proper context subhypergraph is contained in one of these deletions, and removing further context equations preserves a violating assignment. Any triple of pairwise nonorthogonal rays exclusive on a hypergraph with fewer contexts would also be exclusive on the full hypergraph, hence would have to be the same hub triple. This proves the minimum by 13 deletion witnesses and monotonicity. Direct filtering of the 25 and 21 original states gives exactly the 10 and 6 extendable states specified above. □

These filters are statements about binary state families. Abstract Specker-bug attachments can enforce the listed inequalities, because each allowed tip assignment $(0,0), (0,1), (1,0)$ extends to a bug state. A faithful three-dimensional realization of all simultaneous attachments would require a separate check of ray injectivity and unintended orthogonalities. No such geometric extension is asserted here.

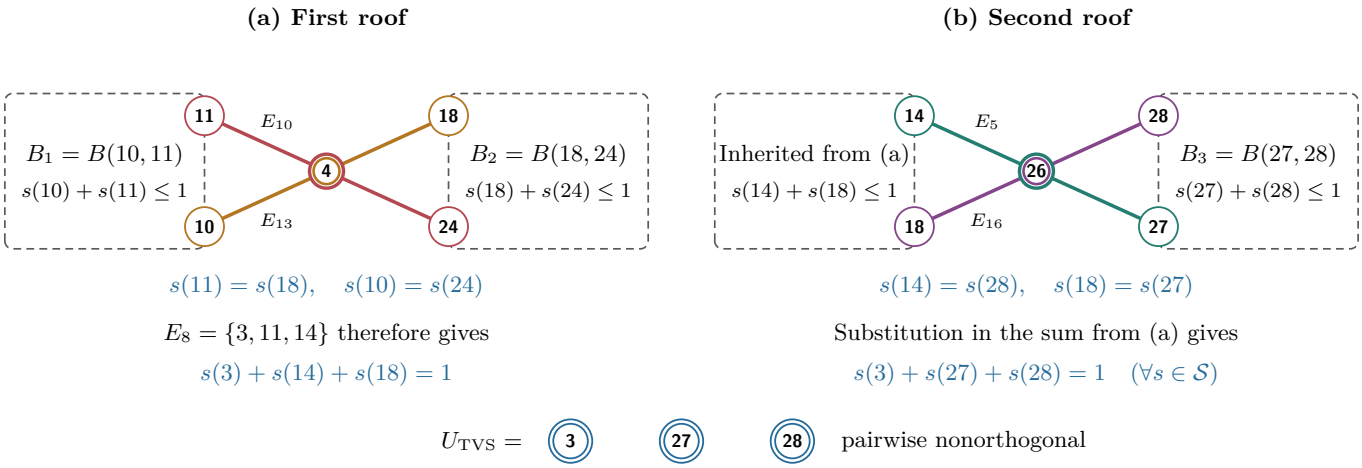


FIG. 6. The two applications of the roof-twin lemma in Theorem 15. Each solid trace contains one complete three-vertex roof context; their shared numbered circle is the roof vertex. Dashed boxes compress seven-context bugs, except the left box in (b), which records the inequality inherited from (a). Box boundaries are not contexts, and internal bug vertices are omitted. In (a), $\{4, 11, 24\}$ and $\{4, 10, 18\}$ force the first twins; context $E_8 = \{3, 11, 14\}$ then gives the sum shown below. In (b), $\{26, 14, 27\}$ and $\{26, 18, 28\}$ force the second twins, yielding normalization on U_{TVS}. The three separate double-outlined markers identify that target triple without adding a context or orthogonality relation. Repeated numbers in the panels refer to the same rays. This is a logical schematic common to $G_{41,25}$ and $G_{41,26}$; the extra context $\{4, 5, 105\}$ of the latter is omitted. The proof and coordinate certificate establish the full faithful geometry independently of this drawing.

G. Color-Weighted Maximal Operators for the Roof-Twin Cascades

Normalize the colors on $U_{\mathrm{TVS}} = (3, 27, 28)$ to $(0, 1, 2)$. For the exact FOR of $G_{41,26}$, the three projections give

$$O_{\mathrm{TVS}} = \frac{c_0}{2}\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{pmatrix} + \frac{c_1}{210}\begin{pmatrix} 64 & -40 & -88 \\ -40 & 25 & 55 \\ -88 & 55 & 121 \end{pmatrix} + \frac{c_2}{210}\begin{pmatrix} 169 & 65 & 52 \\ 65 & 25 & 20 \\ 52 & 20 & 16 \end{pmatrix}. \tag{100}$$

Its full eigensystem follows from the cubic with

$$\begin{aligned} t &= c_0 + c_1 + c_2, \\ b &= \tfrac{32}{35}c_0c_1 + \tfrac{419}{420}c_0c_2 + \tfrac{35}{36}c_1c_2, \\ \delta &= \tfrac{8}{9}c_0c_1c_2. \end{aligned} \tag{101}$$

In particular, the requested parameterized maximum is

$$\max_{\rho}\left[c_0 p_\rho(3) + c_1 p_\rho(27) + c_2 p_\rho(28)\right] = F(t, b, \delta). \tag{102}$$

For unit color values the eigenvalues are approximately 0.68238324, 0.95827730, and 1.35933946, the roots of $630\lambda^3 - 1890\lambda^2 + 1817\lambda - 560 = 0$. A maximizing unit vector is approximately $(-0.63118178, -0.28668161, 0.72071021)^{\mathsf{T}}$, whereas every two-valued state gives total one.

The separate FOR of $G_{41,25}$ must be used for that hypergraph. Let $v_0 = r_3$, $v_1 = r_{27}$, and $v_2 = r_{28}$ be its cross-product coordinates at $(\alpha, \beta) = (9/32, 24/11)$, as constructed above. An exact equality for this representation is again Eq. (102), now with

$$\begin{aligned} b &= \sum_{0 \le i < j \le 2}\left(1 - \frac{|v_i \cdot v_j|^2}{\|v_i\|^2\|v_j\|^2}\right)c_i c_j, \\ \delta &= \frac{2052543025}{2142869041}c_0c_1c_2. \end{aligned} \tag{103}$$

All coefficients are rational and fixed by the stated cross products. At unit color values this operator has maximum 1.21074603. Its optimal eigenspace, and those of $G_{41,26}$ for arbitrary real color values, are exactly the spectral projections in Eq. (18). The same calculation applies to the Yu–Oh targets after the binary restrictions in Proposition 16, since restricting classical states does not alter their already specified quantum projectors.

H. Classical Equalities and Their Spectral Violations in the Cascades

The roof-twin proof gives the exact classical envelope $[\min_j c_j, \max_j c_j]$ for U_{TVS} on all states of either 41-ray hypergraph. At unit values the classical expectation is exactly one, while the quantum intervals are

$$\begin{aligned} G_{41,25}: &\quad [0.79852694, 1.21074603], \\ G_{41,26}: &\quad [0.68238324, 1.35933946]. \end{aligned} \tag{104}$$

Both endpoints violate that normalization. These comparisons use the separate FORs required by the two hypergraphs.

Already in $G_{27,16}$, the partially nonorthogonal triple $\{3, 14, 18\}$ is normalized on every two-valued state. Since $\{3, 11, 14\}$ is an actual context, its unit operator is $I_3 + P_{18} - P_{11}$. In the rational-parameter FOR,

$$\mathrm{spec}(P_3 + P_{14} + P_{18}) = \{1 - \epsilon, 1, 1 + \epsilon\}, \qquad \epsilon = \sqrt{\frac{90326016}{2142869041}} \approx 0.20530930. \tag{105}$$

This supplies a full-state separation even though $G_{27,16}$ has no independent admissibility triple.

The twin differences give another direct comparison. Whenever $s(x) = s(y)$ on all two-valued states, the classical weighted difference $c_j[s(x) - s(y)]$ is identically zero. For two distinct rays its quantum operator has spectrum

$$\{0,\ \pm c_j\sqrt{1 - |\langle \hat{q}_x, \hat{q}_y\rangle|^2}\} \tag{106}$$

in dimension three. Every nonzero c_j therefore produces a separation on an extremal eigenstate. The calculation tests all twin differences separately; a chromatic twin relation is not assumed to be a full-state equality unless the enumerated state restrictions establish it.

VI. FOUR-DIMENSIONAL CHROMATIC RELATIONS IN $\mathbb{R}^4$

In dimension $d = 4$, related chromatic relations occur both in common-apex lifts and in strictly linear 4-uniform hypergraphs. The lifts $B^{(4)}_{14,7}$ and $D^{(4)}_{17,9}$ prepend a singleton apex column to every Context–Color Matrix [9, Proposition 2]; they preserve the three-dimensional universally rainbow pairs and color-multiset-equivalent pairs. The strictly linear examples ($|e \cap e'| \le 1$, girth $g \ge 3$) are $L^{(4)}_{18,7}$ ($|V| = 18$, $|E| = 7$, $N_c = 144$, $|\mathcal{S}_{\mathrm{ext}}|/|\mathcal{S}| = 68/68$) and $L^{(4)}_{22,9}$ ($|V| = 22$, $|E| = 9$, $N_c = 216$, $|\mathcal{S}_{\mathrm{ext}}|/|\mathcal{S}| = 92/158$) [8, 9]. The complete rainbow quadruples exhibited below are partially nonorthogonal, rather than independent admissibility contexts.

In $L^{(4)}_{22,9}$ (Fig. 7), the 9 four-element contexts are

$$\begin{aligned} C_1 &= \{1, 2, 4, 3\}, & C_2 &= \{5, 8, 7, 6\}, \\ C_3 &= \{4, 9, 10, 8\}, & C_4 &= \{11, 1, 12, 5\}, \\ C_5 &= \{11, 14, 13, 2\}, & C_6 &= \{15, 3, 16, 7\}, \\ C_7 &= \{15, 17, 18, 6\}, & C_8 &= \{11, 21, 22, 19\}, \\ C_9 &= \{13, 20, 15, 19\}, \end{aligned} \tag{107}$$

with exact integer FOR in $\mathbb{R}^4$ given by

$$\begin{aligned} r_1 &= (1, 0, 0, 0), & r_2 &= (0, 1, 0, 0), \\ r_3 &= (0, 0, 1, 0), & r_4 &= (0, 0, 0, 1), \\ r_5 &= (0, 1, -1, -1), & r_6 &= (4, -5, -11, 6), \\ r_7 &= (1, -4, 0, -4), & r_8 &= (4, 1, 1, 0), \\ r_9 &= (1, -5, 1, 0), & r_{10} &= (2, -1, -7, 0), \\ r_{11} &= (0, 0, 1, -1), & r_{12} &= (0, 2, 1, 1), \\ r_{13} &= (1, 0, 4, 4), & r_{14} &= (8, 0, -1, -1), \\ r_{15} &= (4, 2, 0, -1), & r_{16} &= (4, -5, 0, 6), \\ r_{17} &= (5, -9, 7, 2), & r_{18} &= (2, 113, 77, 234), \\ r_{19} &= (16, -33, -2, -2), & r_{20} &= (4, 2, -21, 20), \\ r_{21} &= (22, 12, -11, -11), & r_{22} &= (129, 44, 153, 153). \end{aligned} \tag{108}$$

Across all $N_c = 216$ normalized 9×4 Context–Color Matrices of $L^{(4)}_{22,9}$, the base subhypergraph $L^{(4)}_{18,7}$ $(C_1, \ldots, C_7)$ forces the universally rainbow pair $c(11) \neq c(15)$. In $C_5 = \{11, 14, 13, 2\}$ and $C_9 = \{13, 20, 15, 19\}$, this propagates to 7 further nonorthogonal rainbow pairs

$$\begin{aligned} &\{11, 15\},\ \{1, 20\},\ \{2, 20\},\ \{5, 20\}, \\ &\{12, 20\},\ \{14, 20\},\ \{15, 21\},\ \{15, 22\}, \end{aligned} \tag{109}$$

the monochromatic nonorthogonal pair $11 \sim_c 20$ ($c(11) = c(20)$), and 4 partially nonorthogonal complete rainbow quadruples ($\{c(u_1), c(u_2), c(u_3), c(u_4)\} = \{0, 1, 2, 3\}$ in all 216 normalized colorings):

$$\begin{aligned} &\{19, 20, 21, 22\}, \quad \{1, 5, 12, 20\}, \\ &\{2, 13, 14, 20\}, \quad \{11, 13, 15, 19\}. \end{aligned} \tag{110}$$

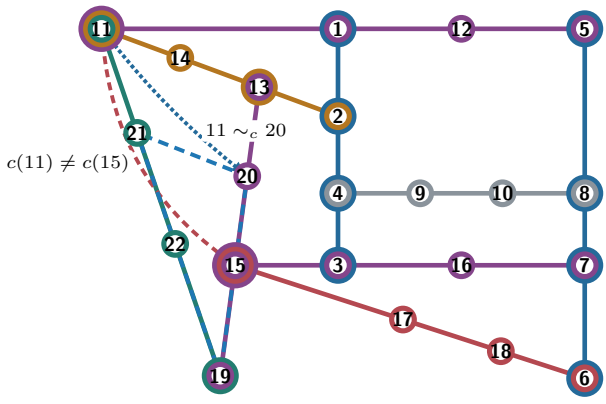


FIG. 7. The strictly linear 22-ray, 9-context hypergraph $L^{(4)}_{22,9}$ in $\mathbb{R}^4$ [Eqs. (107)–(108)], with all 9 four-element contexts drawn as straight lines. Shown are the universally rainbow pair $c(11) \neq c(15)$, the monochromatic pair $11 \sim_c 20$, and the complete rainbow quadruple $\{19, 20, 21, 22\}$ (blue dashed polygon).

TABLE IV. Census of chromatic and two-valued state admissibility contexts and related relations across 19 quantum hypergraphs in $\mathbb{R}^3$, $\mathbb{C}^3$, and $\mathbb{R}^4$. Listed are the Hilbert space $\mathbb{F}^d$, vertex/context counts, normalized coloring count N_c, extendable-to-total state ratio, universally rainbow nonorthogonal pairs, chromatic-twin pairs, chromatic admissible contexts on $\mathcal{S}_{\text{ext}}(H)$ (with additional partially nonorthogonal complete rainbow d-tuples in parentheses), two-valued state admissibility contexts on $\mathcal{S}(H)$ (with additional partially nonorthogonal two-valued complete d-tuples in parentheses), and non-twin color-multiset-equivalent pairs of sizes $k = 2, 3$. The paired columns omit pairs of two already complete admissible contexts; both independent triples of $H_{53,34}$ are normalized on all two-valued states.

Config.	$\mathbb{F}^d$	$(\lvert V\rvert, \lvert E\rvert)$	N_c	$\lvert\mathcal{S}_{\text{ext}}\rvert/\lvert\mathcal{S}\rvert$	Rainbow pairs	Twin pairs	Chromatic admissibility	Two-valued admissibility	Paired $k=2$	Paired $k=3$
$M_{15,8}$	$\mathbb{R}^3$	$(15, 8)$	16	24/24	0	0	0 (0)	0 (0)	0	4
H_{20}	$\mathbb{C}^3$	$(20, 12)$	12	26/26	0	0	0 (0)	0 (0)	1	20
$B_{13,7}$	$\mathbb{R}^3$	$(13, 7)$	6	12/14	1	0	0 (0)	0 (0)	0	1
$D_{16,9}$	$\mathbb{R}^3$	$(16, 9)$	7	14/21	1	0	0 (0)	0 (0)	0	0
$I_{16,9}$	$\mathbb{R}^3$	$(16, 9)$	6	12/22	10	2	0 (6)	0 (0)	2	3
$H_{17,10}$	$\mathbb{R}^3$	$(17, 10)$	4	10/18	16	2	0 (6)	0 (0)	2	3
$T_{19,11}$	$\mathbb{R}^3$	$(19, 11)$	7	14/25	8	2	0 (4)	0 (0)	1	2
$H_{20,12}$	$\mathbb{R}^3$	$(20, 12)$	4	10/25	28	4	**2** (12)	0 (0)	8	24
$H_{21,13}$	$\mathbb{R}^3$	$(21, 13)$	2	6/21	66	21	**21** (63)	0 (0)	36	0
$G_{27,16}$	$\mathbb{R}^3$	$(27, 16)$	36	54/82	18	2	0 (10)	0 (10)	4	10
$G_{41,25}$	$\mathbb{R}^3$	$(41, 25)$	216	216/396	46	6	**1** (29)	**1** (29)	10	29
$G_{41,26}$	$\mathbb{R}^3$	$(41, 26)$	48	63/107	60	6	**1** (29)	**1** (29)	10	308
$H_{53,34}$	$\mathbb{R}^3$	$(53, 34)$	128	128/224	66	4	**2** (23)	**2** (23)	12	416
$R_{21,12}$	$\mathbb{C}^3$	$(21, 12)$	32	48/54	0	0	0 (0)	0 (0)	0	12
$R_{25,15}$	$\mathbb{C}^3$	$(25, 15)$	24	48/81	0	0	0 (0)	0 (0)	0	12
$B^{(4)}_{14,7}$	$\mathbb{R}^4$	$(14, 7)$	6	13/15	1	0	0 (0)	0 (0)	0	1
$D^{(4)}_{17,9}$	$\mathbb{R}^4$	$(17, 9)$	7	15/22	1	0	0 (0)	0 (0)	0	0
$L^{(4)}_{18,7}$	$\mathbb{R}^4$	$(18, 7)$	144	68/68	1	0	0 (0)	0 (0)	0	0
$L^{(4)}_{22,9}$	$\mathbb{R}^4$	$(22, 9)$	216	92/158	8	1	0 (4)	0 (0)	0	0

A. Four-Parameter Spectral Equalities and Maximal Preparations

All four complete rainbow quadruples in Eq. (110) arise by exchanging the chromatic twins 11 and 20 in an actual context: the first three replace 11 by 20 in C_8, C_4, C_5, respectively, and the last replaces 20 by 11 in C_9. If C is that context and $t = \kappa(11) = \kappa(20)$, their weighted operators are therefore

$$O_U(\mathbf{c};\kappa) = A_C(\mathbf{c};\kappa) \ \pm c_t(P_{20} - P_{11}), \tag{111}$$

where the plus sign applies to the first three. This constructs the operator from an actual spectral sum followed by a rank-at-most-two Hermitian correction.

For an explicit four-parameter equality, order the first target as $U = (20, 19, 21, 22)$ and label its colors $(0, 1, 2, 3)$. The vector $\hat{r}_{19}$ is orthogonal to the other three target rays, so it is an eigenvector with eigenvalue c_1. The squared overlaps needed on its orthogonal complement are

$$\begin{aligned} |\langle \hat{r}_{20}, \hat{r}_{11}\rangle|^2 &= \tfrac{41}{42}, \quad & |\langle \hat{r}_{20}, \hat{r}_{19}\rangle|^2 &= 0, \\ |\langle \hat{r}_{20}, \hat{r}_{21}\rangle|^2 &= \tfrac{41}{2030}, \quad & |\langle \hat{r}_{20}, \hat{r}_{22}\rangle|^2 &= \tfrac{11}{3045}. \end{aligned} \tag{112}$$

The characteristic equality factors as

$$\begin{aligned} 0 &= (\lambda - c_1)\big(\lambda^3 - T\lambda^2 + B\lambda - K\big), \\ T &= c_0 + c_2 + c_3, \\ B &= \tfrac{1989}{2030}c_0c_2 + \tfrac{3034}{3045}c_0c_3 + c_2c_3, \\ K &= \tfrac{41}{42}c_0c_2c_3. \end{aligned} \tag{113}$$

Thus the exact maximal expectation is

$$\boxed{\max_\rho \big[c_0 p_\rho(20) + c_1 p_\rho(19) + c_2 p_\rho(21) + c_3 p_\rho(22)\big] = \max\{c_1, F(T, B, K)\}.} \tag{114}$$

The cubic formula gives the other three eigenvalues; Eq. (18) gives the full eigenspaces, including any degeneracy with c_1. The other quadruples follow from Eq. (111) or directly from the general principal-minor equality.

For unit color values, every context operator is I_4, so all four target operators have the same spectrum,

$$\operatorname{spec} O_U(\mathbf{1}) = \{1 - 1/\sqrt{42},\ 1,\ 1,\ 1 + 1/\sqrt{42}\}. \tag{115}$$

Their maximal expectation is exactly $1+1/\sqrt{42}$, compared with the chromatic state-sum value one. For an explicit

maximizing preparation for the first three targets, put $u = \hat{r}_{20}$, $v = \hat{r}_{11}$, $\alpha = \langle v, u\rangle$, and $\varepsilon = 1/\sqrt{42}$. A vector proportional to

$$u - \frac{\alpha}{1+\varepsilon} v$$

is an eigenvector of $P_{20} - P_{11}$ with eigenvalue ε; normalize it to obtain the maximizing pure state. For the fourth target, exchange u and v. The double eigenvalue one has eigenspace $\mathrm{span}\{u, v\}^{\perp}$.

Finally, a common-apex lift sends a three-dimensional target operator to $O_X \oplus 0$, adding a zero eigenvalue. If the apex is included, its color-weighted projection changes this to $O_X \oplus c_{\kappa(\mathrm{apex})}$. Hence the lifted maxima are $\max\{\lambda_{\max}(O_X), 0\}$ and $\max\{\lambda_{\max}(O_X), c_{\kappa(\mathrm{apex})}\}$, respectively. The rainbow pairs of $L^{(4)}_{18,7}$ and $L^{(4)}_{22,9}$ have the two eigenvalues in Eq. (81) together with two zeros. These constructions cover the four-dimensional census using at most four independent real color parameters.

B. Full Classical Bounds in Four Dimensions

In $L^{(4)}_{22,9}$, the full two-valued state family satisfies the one-way implication $s(11) \le s(20)$. Indeed, $s(11) = 1$ sets $s(13) = s(19) = 0$, the $\{11, 15\}$ TIFS gives $s(15) = 0$, and C_9 then forces $s(20) = 1$. The reverse implication fails on 66 of the 158 states. Thus the first three quadruples of Eq. (110) have totals $1 + s(20) - s(11) \in \{1, 2\}$, while the last has total $1 + s(11) - s(20) \in \{0, 1\}$. Their common quantum spectrum in Eq. (115) violates the lower classical bound for the first three and the upper bound for the last.

More generally, in the color order $(20, 19, 21, 22)$ of Section VI A, the exact classical four-parameter endpoints are the minimum and maximum of

$$\{c_0, c_1, c_2, c_3, c_0 + c_2, c_0 + c_3\}. \tag{116}$$

These are the bounds to compare with the four quantum roots in Eq. (113). On the extendable states, only the four single-color expressions remain. The twin difference $s(11) - s(20)$ has full classical interval $[-1, 0]$, whereas $P_{11} - P_{20}$ has extrema $\pm 1/\sqrt{42}$.

The pair $\{11, 15\}$ is already exclusive on all 68 states of $L^{(4)}_{18,7}$, so its quantum maximum $1 + 1/\sqrt{42}$ exceeds the full classical bound one. The common-apex lifts preserve the corresponding base comparisons, with the additional classical possibility of selecting only the apex. Consequently the bug-tip separation persists in $B^{(4)}_{14,7}$, whereas the pair $\{14, 18\}$ in $D^{(4)}_{17,9}$ still separates only the extendable-state bound. The full inventories use the separate faithful coordinates for $L^{(4)}_{18,7}$ and $L^{(4)}_{22,9}$, including their different rays 13 and 14.

VII. CLASSICAL–QUANTUM COMPARISON ACROSS ALL CONFIGURATIONS

The comparison covers every configuration in Table IV. The accompanying Python inventory explicitly specifies the FOR used for each graph and reconstructs every listed context. Complete state enumeration is followed by projection to the relevant color coefficient vectors, so the classical bounds remain exact functions of arbitrary real color values. Table V illustrates one target per configuration at unit values; the complete machine-readable comparison contains every target expression described below.

The exhaustive target inventory contains 2,289 single-set sums, 931 signed size-two or size-three paired differences, and 50 twin differences: 3,270 expressions in total. Single-set sums include actual contexts as consistency checks, all non-context complete rainbow sets, independent universally rainbow subsets, nontrivial twin subsets, and both components of every counted paired relation. Paired relations use the census convention excluding pairs of two already complete independent admissibility contexts. Restricting every normalized strong coloring to each expression, and identifying restrictions related by a palette permutation, gives 10,873 color patterns. This covers every coloring and every real assignment to its used colors; it does not enumerate arbitrary unlisted subsets of vertices.

For every pattern, the saved integer vectors define both classical envelopes in Eq. (27). Testing all classical facets and both signs of the affine equalities gives the comparison counts in Table VI. Unit values already separate 663 expressions from the full state family; allowing general real color values increases this number to 1,445. These are overlapping target expressions, not 1,445 independent contextuality proofs. A zero entry rules out separation within the listed color-weighted expression family and chosen FOR to the reported numerical precision; it does not establish a classical model for other observables or other FORs.

A. Enumeration and Numerical Verification

The supplied Python translation of Tkadlec's program [13] agrees, state by state, with the new block-recursion implementation and an independent variable-first propagation solver for all 19 configurations. A further 240 small complete-block/proper-subblock problems agree with exhaustive binary enumeration. No state-generation discrepancy was found for the covered vertex sets. The legacy Pascal executable could not be launched on the verification host, so no Pascal execution or compilation is claimed. The new module ports the state-generation algorithm, not the original potentially incomplete element-separation and order-determination heuristics; those heuristics are unnecessary for Eq. (27). The original source and supplied translation are preserved

TABLE V. Representative unit-color comparisons for all 19 configurations. Classical entries are sharp expectation intervals over convex mixtures of all two-valued states ($\mathcal{S}$) or only extendable states ($\mathcal{S}_{\text{ext}}$). Quantum entries are the extremal eigenvalues for the specified FOR, rounded to six decimal places. The last column identifies separation from the full classical interval, not merely from the extendable-state interval. The selected row for a configuration need not maximize its separation. The accompanying tables retain every expression.

Configuration	Target	Classical on $\mathcal{S}$	Classical on $\mathcal{S}_{\text{ext}}$	Quantum interval	Separation
$M_{15,8}$	$\{1, 6, 11\}$	$[0, 2]$	$[0, 2]$	$[0.172673, 2.000000]$	none
H_{20}	$\{a_1, a_2\}$	$[0, 2]$	$[0, 2]$	$[0.000000, 1.553775]$	none
$B_{13,7}$	$\{10, 11\}$	$[0, 1]$	$[0, 1]$	$[0.000000, 1.333333]$	upper
$D_{16,9}$	$\{14, 18\}$	$[0, 2]$	$[0, 1]$	$[0.000000, 1.088889]$	none
$I_{16,9}$	$\{10, 11\}$	$[0, 1]$	$[0, 1]$	$[0.000000, 1.272166]$	upper
$H_{17,10}$	$\{10, 11\}$	$[0, 1]$	$[0, 1]$	$[0.000000, 1.333333]$	upper
$T_{19,11}$	$\{9, 26\}$	$[0, 1]$	$[0, 1]$	$[0.000000, 1.252415]$	upper
$H_{20,12}$	$\{15, 18, 20\}$	$[1, 3]$	$[1, 1]$	$[0.729610, 1.234838]$	lower
$H_{21,13}$	$\{10, 11, 13\}$	$[0, 1]$	$[1, 1]$	$[0.333333, 1.333333]$	upper
$G_{27,16}$	$\{3, 14, 18\}$	$[1, 1]$	$[1, 1]$	$[0.794691, 1.205309]$	both
$G_{41,25}$	$\{3, 27, 28\}$	$[1, 1]$	$[1, 1]$	$[0.798527, 1.210746]$	both
$G_{41,26}$	$\{3, 27, 28\}$	$[1, 1]$	$[1, 1]$	$[0.682383, 1.359339]$	both
$H_{53,34}$	$\{a, b, u\}$	$[1, 1]$	$[1, 1]$	$[0.570956, 1.516521]$	both
$R_{21,12}$	$\{1, 17, 18\}$	$[0, 3]$	$[0, 3]$	$[0.500000, 1.762348]$	none
$R_{25,15}$	$\{1, 17, 18\}$	$[0, 3]$	$[0, 3]$	$[0.500000, 1.762348]$	none
$B^{(4)}_{14,7}$	$\{10, 11\}$	$[0, 1]$	$[0, 1]$	$[0.000000, 1.333333]$	upper
$D^{(4)}_{17,9}$	$\{14, 18\}$	$[0, 2]$	$[0, 1]$	$[0.000000, 1.088889]$	none
$L^{(4)}_{18,7}$	$\{11, 15\}$	$[0, 1]$	$[0, 1]$	$[0.000000, 1.154303]$	upper
$L^{(4)}_{22,9}$	$\{19, 20, 21, 22\}$	$[1, 2]$	$[1, 1]$	$[0.845697, 1.154303]$	lower

with the calculation.

The legacy programs assume that every declared atom occurs in a listed block or subblock. An unused declared atom is silently fixed to zero; the new implementation instead enumerates it freely. This difference does not affect the present hypergraphs, for which $V = \bigcup E$. The new program also removes the fixed array limits and writes results separately rather than overwriting its input. Every configuration has a dense Pascal-format input/state file together with an explicit map back to the manuscript's ray labels.

All state, extendability, and normalized-coloring counts agree with the manuscript census. The only relation-count correction is the octagon's 20 size-three affine pairs, with an explicit certificate in Eq. (49). For every chosen FOR, all contexts resolve the identity, all intended orthogonal pairs are verified, all nonedges have nonzero inner product, and all rays are distinct. The separate coordinates for deleted-context restrictions, roof cascades, and four-dimensional examples are retained; a nonfaithful restriction of a larger FOR is not substituted for them.

There are 218 distinct classical coefficient polytopes. Their integer facet planes are reconstructed exactly and checked against every state coefficient vector; an independent enumeration of all possible supporting hyperplanes reproduces every facet list. Quantum eigenvalues for 20,527 distinct signed operators are independently recomputed at 80 decimal digits, verifying all 100,495 facet/equality comparisons and every unit expression. The largest discrepancy from the working double-precision extrema is below 3.1×10^{-15}. Reported strict separations exceed 10^{-9}; the smallest positive gap in the scan is greater than 7.2×10^{-4}. Maximizing vectors and their residuals are saved with the witnesses. These are numerical spectral comparisons with exact classical bounds, not a global optimization of gap magnitude over coefficients or FORs.

The files `unit-comparison.csv` and `weighted-witnesses.csv` give all unit expressions and separating parameter choices. Each configuration's JSON record contains its complete states and colorings, every distinct restricted color pattern, both exact finite classical envelopes, affine equalities, facet inequalities, and spectral comparisons. The evaluator accepts arbitrary real color values and can construct the exact symbolic operator and characteristic polynomial from the algebraic ray formulas. Thus the calculation is reproducible without the missing certificate referenced by the older verifier.

VIII. MASTER CENSUS AND CONCLUSION

Table IV summarizes the invariants and admissibility relations of all 19 configurations, including $H_{53,34}$. Its two independent chromatic admissible triples, $\{a, b, u\}$ and

TABLE VI. Coverage and separation counts for the full comparison. The triple $X/D/T$ counts single-set sums, multiset-pair differences, and twin differences, respectively. $U_{\mathcal{S}}$ counts expressions separated from $\operatorname{conv}\mathcal{S}$ at unit color values. $W_{\mathcal{S}}$ and W_{ext} count expressions admitting a separating real color assignment for at least one restricted color pattern, using the complete facet/equality test on the indicated state family. Both spectral endpoints are included. Actual contexts are part of X but never contribute a separation. Counts depend on the stated FORs.

Configuration	$\lvert\mathcal{S}_{\text{ext}}\rvert/\lvert\mathcal{S}\rvert$	$X/D/T$	$U_{\mathcal{S}}$	$W_{\mathcal{S}}$	W_{ext}
$M_{15,8}$	24/24	16/4/0	0	0	0
H_{20}	26/26	54/21/0	0	0	0
$B_{13,7}$	12/14	10/1/0	1	1	1
$D_{16,9}$	14/21	10/0/0	0	0	1
$I_{16,9}$	12/22	37/5/2	13	13	26
$H_{17,10}$	10/18	44/5/2	13	13	32
$T_{19,11}$	14/25	31/3/2	2	2	17
$H_{20,12}$	10/25	110/32/4	21	38	117
$H_{21,13}$	6/21	199/36/21	30	45	207
$G_{27,16}$	54/82	74/14/2	42	46	50
$G_{41,25}$	216/396	173/39/6	120	139	146
$G_{41,26}$	63/107	571/318/6	186	484	620
$H_{53,34}$	128/224	841/428/4	227	656	844
$R_{21,12}$	48/54	33/12/0	0	0	0
$R_{25,15}$	48/81	36/12/0	0	0	0
$B^{(4)}_{14,7}$	13/15	10/1/0	1	1	1
$D^{(4)}_{17,9}$	15/22	10/0/0	0	0	1
$L^{(4)}_{18,7}$	68/68	8/0/0	1	1	1
$L^{(4)}_{22,9}$	92/158	22/0/1	6	6	13

$\{a', b', u\}$, also satisfy the stronger two-valued state admissibility condition. The paired columns omit equivalences between two already complete admissibility contexts, as specified in Definition 2.

The central phenomenon is enforced exclusivity and completeness on a single independent set, stratified into a three-tier hierarchy by the underlying state family:

1. *Chromatic admissible contexts on $\mathcal{S}_{\text{ext}}(H)$:* Rainbow in every strong d-coloring, or equivalently normalized ($\sum_{u\in U} s(u) = 1$) on every extendable color-class state. In the Yu–Oh family, pairwise-nonorthogonal chromatic admissibility triples first appear at 12 contexts ($H_{20,12}$, 2 triples) and reach 21 triples in $H_{21,13}$ (Theorem 12), including the coplanar triple $U_{\text{cop}} = \{3, 6, 9\}$ and the 60° basis $U_{\text{core}} = \{1, 4, 7\}$.

2. *Two-valued admissibility contexts on $\mathcal{S}(H)$:* Normalized ($\sum_{u\in U} s(u) = 1$) on *every* admissible two-valued state $s \in \mathcal{S}(H)$. Even though no linear combination of context equations can force a nonorthogonal d-tuple to sum to 1 on $\mathcal{A}(H)$ (Proposition 9), $\{0, 1\}$-integrality on Specker bugs across two-context roofs enforces biconditional two-valued twins $s(x) = s(x')$ (Lemma 13), causing $\operatorname{aff}\mathcal{S}(H) \subsetneq \mathcal{A}(H)$ and yielding the pairwise-nonorthogonal two-valued state admissibility context $U_{\text{TVS}} = \{3, 27, 28\}$ in $G_{41,25}$ and $G_{41,26}$ (Theorem 15). Proposition 16 gives additional binary state restrictions for $H_{20,12}$ and $H_{21,13}$; faithful geometric realizations of simultaneous gadget attachments are not claimed. The 53-ray construction of Section IV gives two further independent two-valued state admissibility contexts and an explicit Born-state total 8/9, with a full hypergraph drawing and exact coordinate certificate.

3. *Operator and affine rigidity on $\mathcal{W}(H)$ and $\mathcal{A}(H)$:* No nonorthogonal d-tuple can resolve the identity $\sum_{u\in U} P_u = I_d$ or be normalized on $\mathcal{A}(H)$. Meanwhile, balanced context certificates yield paired affine pseudocontexts ($M_{15,8}$ and H_{20}), which imply color-multiset equivalence on $\mathcal{S}_{\text{ext}}(H)$ without requiring single-set completeness.

The color-weighted construction adds a quantitative quantum comparison to these assignment-level distinctions. For any fixed FOR and coloring, Eqs. (20) and (19) give the characteristic equality and the largest attainable expectation as functions of the used real color values. The resulting eigenspaces identify optimal preparations; a chromatic or two-valued normalization does not replace this spectral calculation.

The full-state comparisons also distinguish spectral violations of upper and lower bounds. In particular, the $H_{20,12}$ triples violate the full classical lower bound, whereas the core and coplanar triples of $H_{21,13}$ remain

inside their full projected classical polytopes. The hub triple, roof cascades, pasted construction, and selected four-dimensional relations give full-state separations. Tables V and VI record the comparison for all configurations and specify its observable and representation scope.

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